

Can Social-Welfare Objectives Improve Distributional Fairness in Multi-Agent Reinforcement Learning?

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Author's Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, the work is the candidate's own work. Work done in collaboration with, or with the assistance of, others, is indicated as such. Any views expressed in the dissertation are those of the author.

A handwritten signature in black ink, reading "Li Shen". The signature is written in a cursive style, with the first letter of each name being capitalized and prominent.

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10 September 2026

Executive Summary

Multi-agent reinforcement learning (MARL) can coordinate a highway merge without necessarily distributing the cost of coordination fairly. This dissertation studies that problem in a simulated four-vehicle merge with Fast and Slow vehicles on both the ramp and mainline. Because the two classes have different mobility requirements, welfare is measured relative to each vehicle’s own target speed rather than inferred from visible actions such as braking.

The study proceeds through a matched experimental sequence. EXP01 characterises the task-only Baseline welfare distribution. EXP02 compares Baseline with Mean, Generalized Gini (GGI), and Rawlsian Maximin terminal social-welfare objectives across 12 independently trained seeds. Welfare-State Communication (WSC) tests whether policies use explicit welfare history, Dense Welfare Shaping (DWS) tests earlier welfare feedback, and EXP05 combines accumulated burden, visible-neighbour comparison, and action-dependent local feedback while retaining terminal Rawlsian Maximin.

Baseline results show that visible concession and experienced disadvantage differ: Slow vehicles brake more often, whereas Fast vehicles are more frequently worst-off and carry greater target-relative mobility burden. In EXP02, Mean and GGI move some welfare measures favourably on average, but the seed-level confidence intervals cross zero; Rawlsian Maximin also shows no reliable improvement. WSC provides stronger mechanism evidence: trained policies use welfare-history inputs, and targeted yielding is followed by substantially more local welfare-gap recovery. DWS shows that earlier feedback does not by itself resolve temporal and local credit assignment. EXP05 produces a favourable mean paired change in overall worst-off welfare of +0.114 at 3.8M steps, with 8 of 12 seeds moving upward, but the 95% interval $[-0.086, +0.294]$ includes zero and clear within-success gains occur in only two seeds.

The results do not support the claim that changing a social-welfare objective is sufficient on its own. Instead, they identify where fairness information influences behaviour and where that influence is lost between local decisions and final welfare outcomes.

Main contributions and achievements are:

- I developed a target-relative mobility-welfare framework that separates physical concession from experienced burden in a heterogeneous merge task (Chapters 3–4).
- I conducted a 12-seed matched comparison of Baseline, Mean, GGI, and Rawlsian Maximin, using training seed as the unit of uncertainty (chapter 4).
- I used WSC and DWS to distinguish information availability from temporal and agent-level credit assignment (chapter 4).

- I designed and critically evaluated an EXP05 follow-up combining accumulated burden, local comparison, and action-dependent feedback (Chapters 4–6).

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Finally, I would like to thank my friends who stood by me during difficult times.

Statement on the inclusion of published and collaborative work

My dissertation does not include work that has been published, submitted or prepared for publication with me as an author. My dissertation does not include non-published collaborative work that contains data or contributions from others.

The information in this statement counts as the references to my own publications if they are used as whole chapters or as whole sections of chapters. If work from my own publications have been incorporated in smaller segments, I have included individual references to them in the relevant part of the dissertation in addition to completing any required table.

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1. Introduction

1.1. Motivation

Highway merging is an everyday scenario that requires coordination among multiple vehicles, and the cost of that coordination is not always shared evenly. A ramp vehicle may slow to enter behind mainline traffic, or a mainline vehicle may give way. Both outcomes count as success, but the vehicles do not necessarily pay the same price. That makes highway merging a useful setting for studying fairness in multi-agent reinforcement learning.

Much prior work on highway merging with reinforcement learning targets coordination itself—higher completion rates, better efficiency, lower collision rates [1], [2]. Fewer studies ask which agents repeatedly lose mobility or face greater risk.

This study therefore separates task performance from individual welfare and uses social-welfare objectives—Mean, Generalized Gini (GGI), and Rawlsian Maximin—to influence the final welfare distribution. Mean favours average welfare, GGI weights lower-ranked utilities more heavily, and Rawlsian Maximin focuses on the worst-off vehicle.

Later experiments ask why changing that final objective may still fail to change local behaviour. Welfare-State Communication (WSC) adds welfare history to the observation; Dense Welfare Shaping (DWS) delivers welfare feedback during the episode; and a final follow-up combines accumulated burden with action-dependent local feedback.

The formal task contains four learned vehicles: one Fast and one Slow on the ramp,

1. Introduction

and one Fast and one Slow on the mainline. They share one Deep Q-Network (DQN) but act from separate local observations. The aim is to see how the burden of a learned merge is distributed and whether a social-welfare preference can change that distribution.

1.2. Research Questions

RQ1 — Welfare distribution under task-only coordination

Do systematic inequalities in individual mobility welfare emerge under task-only highway-merging coordination with heterogeneous mobility requirements?

RQ1 is evaluated using worst-off utility, Utility Gini, worst-off identity, and target-relative mobility burden. Because collision, timeout, or incompleteness can reduce final utility to zero, completion and collision are analysed alongside the welfare measures.

RQ2 — Social-welfare objectives

Compared with task-only learning, do Mean, Generalized Gini, and Rawlsian Maximin social-welfare objectives improve worst-off welfare and reduce inequality in realised individual welfare outcomes?

The primary outcomes are worst-off utility $U_{\min}$ and Utility Gini. WSC and DWS are not additional formal research questions; they are diagnostic experiments introduced after the main objective comparison to explain why a group-level welfare preference does or does not reach local behaviour.

1.3. Contributions

The main empirical contribution is the Baseline analysis of who carries the coordination cost. Fast vehicles are more exposed to the welfare lower tail even though Slow

vehicles brake more often, so visible concession and requirement-relative burden are not interchangeable.

The later experiments separate three learning choices. EXP02 changes only the terminal social-welfare objective; WSC changes the information available to the policy; DWS changes when welfare feedback is delivered. The final follow-up combines accumulated burden, local comparison, and action-dependent feedback. It produces clear within-success gains in two policies but no consistent improvement across 12 independent training runs.

Those 12 runs—not individual test episodes—are the unit used to estimate uncertainty.

2. Background

Highway coordination, distributive fairness, and multi-agent reinforcement learning (MARL) intersect throughout this dissertation, and none of the three can be taken for granted on its own. I begin with what prior work has achieved in highway merging, then narrow the focus to fairness. The closing sections introduce the learning machinery that [chapter 3](#) reuses: `highway-env`, Double DQN, parameter sharing, and curriculum learning.

2.1. Reinforcement Learning for Highway Coordination

A highway merge is not a set of independent driving decisions. When one vehicle takes a gap, another loses it. Chen et al. [\[1\]](#) model this interaction as cooperative MARL in mixed on-ramp traffic. Their autonomous vehicles operate on both ramp and mainline; parameter sharing and local rewards keep the controller scalable. The system also masks unsafe or invalid actions, adds a priority-based safety supervisor, and ramps up task difficulty through curriculum learning.

Wang et al. [\[2\]](#) take a different route and train a single DQN rather than a cooperative multi-agent controller. Their reward combines merge completion, safety, and efficiency, and the agent learns useful longitudinal control in changing traffic. Distributional fairness, however, is not what they measure.

Similar difficulties appear across the wider MARL literature. Dinneweth et al. [\[3\]](#)

2. Background

point to partial observability and non-stationary co-learners as recurring problems in autonomous driving. Ji et al. [4] add V2X information and use centralised training with distributed execution to improve cooperative merging; they also analyse ramp and mainline behaviour separately. How well the policies work depends on traffic density.

That literature answers an important first question: can learning produce coordinated merging? In the settings reviewed here, the answer is yes. This thesis starts from the next one. Once coordination works, who repeatedly gives up mobility to make it work?

2.2. Philosophical Perspectives on Fairness and Distributive Justice

Fairness is not the same as identical physical outcomes. Aristotle distinguishes simple equality from proportional treatment [5]. Sen later asks “equality of what?” [6], forcing attention onto the quantity being compared rather than assuming that one common measure always fits.

The merge environment makes that distinction concrete. Fast vehicles target 22 m/s and Slow vehicles target 18 m/s. Two vehicles can therefore travel at different absolute speeds while each still meets its own requirement. I use a target-relative welfare measure for that reason. It remains a modelling choice: Aristotle and Sen help frame the comparison, but they do not establish target-speed attainment as the only valid measure.

Mill [7] offers a different starting point by emphasising aggregate welfare. Mean welfare is the straightforward computational analogue—every vehicle counts, and the group average matters. That simplicity has a cost, because a strong outcome for one vehicle can numerically offset a weak one elsewhere.

Rawls shifts attention toward the least advantaged [8]. The Maximin objective used later borrows only that narrow priority. Rawls’s Difference Principle is much broader than $\min_i U_i$, so the experiments should not be read as a direct implementation of Rawlsian

justice.

Parfit [9] separates equality from priority, and the distinction matters empirically. Utility Gini captures how spread out the four final utilities are; $U_{\min}$ captures how the worst vehicle fared. The two numbers can move in different directions, so I report both.

The philosophy in this chapter sets boundaries on interpretation. It does not, by itself, write the reward function.

2.3. Distributional Fairness in Traffic Coordination

Traffic fairness has been defined around different protected outcomes. He et al. [10] compare cooperative on-ramp strategies and track how traffic benefit and delay are distributed. In later theoretical work [11], the efficiency–fairness trade-off becomes explicit: the ordering that maximises total traffic performance need not allocate delay most evenly, and improving one objective can worsen the other.

Kunibe et al. [12] protect access instead. Their scheduling scheme balances highway outflow with merge-time assignment for ramp and mainline traffic. Li et al. [13] focus on safety risk and use reinforcement learning to manage asymmetric interaction risk under equality- and Maximin-style principles. The word “fairness” therefore covers different objects in different papers.

This thesis protects mobility. Vehicles here coordinate mainly through speed and progress, and the Fast and Slow classes carry different targets, so welfare is evaluated against each vehicle’s own target speed.

Action and experience must be kept apart. **Physical concession** covers visible behaviour such as braking or yielding. **Experienced burden** covers mobility loss relative to a vehicle’s requirement. A Slow vehicle can brake and still remain above 18 m/s; a Fast vehicle may brake less often yet spend longer below 22 m/s. EXP01 measures both.

2.4. Social-Welfare Functions in Sequential Decision-Making

Individual welfare alone does not say how the group should be valued. Siddique, Weng, and Zimmer [14] embed the aggregation rule in the reinforcement-learning objective: their fair multi-objective framework optimises social welfare directly instead of treating fairness as a post-hoc statistic.

Generalized Gini is one way to do this. Weymark [15] developed the Generalized Gini index so that positions in an ordered distribution can receive different weights; lower-ranked outcomes can count more while the rest of the distribution still matters.

Woodgate, Marshall, and Ajmeri [16] push further toward worst-off protection. Their RAWL · E framework operationalises Maximin in norm-learning agents and tests it in simulated harvesting societies, where minimum experience improves relative to comparison societies. Zhang and Shah [17] include the worst-performing agent in sequential resource allocation while retaining concern for group performance. Ju, Ghosh, and Shroff [18] study the same problem when the multi-agent finite-horizon MDP is unknown and develop online and offline fair-RL methods.

Together, these papers show that social welfare can guide learning, not merely evaluate it. They leave a narrower question open: if the environment, observation, policy, and task reward stay fixed, is changing the aggregation rule enough?

2.5. Fairness-Aware Multi-Agent Reinforcement Learning

In practice, fair-MARL methods rarely change only one scalar objective. Jiang and Lu [19] build fairness into the Fair-Efficient Network (FEN), combining an agent-level fair-efficient reward with a hierarchical controller; their agents learn to switch among behavioural sub-policies, so both reward and architecture change.

Zimmer et al. [20] split the network itself: one sub-network represents efficiency and another equity, and the architecture supports welfare functions such as Generalized Gini and lexicographic Maximin. Grupen, Selman, and Lee [21] define team fairness and enforce it through equivariant policies after showing that simple team-reward maximisation can disadvantage individual members.

Dong et al. [22] divide objectives by level. Local agents optimise egoistic policies while a federated controller with broader information handles utilitarian and egalitarian concerns, so the fairness mechanism is partly an information architecture.

Traffic studies repeat the pattern. Siddique, Li, and Cao [23] use Generalized Gini in decentralised traffic-signal control because maximising expected reward can favour some intersections over others. Toghi et al. [24] redesign distributed rewards around a quantitative social preference; in their mixed-autonomy highway-merging experiments, altruistic agents complete more merges and improve traffic flow and safety relative to egoistic agents.

History can matter as well. Hughes et al. [25] make agents sensitive to inequity in accumulated rewards across temporally extended social dilemmas, and cooperation improves in several Markov games—the mechanism is not only a terminal welfare function.

The literature is encouraging, but it complicates attribution. A fairer endpoint may come from the objective, the observation, the network, the reward timing, or their combination. EXP02 is designed to isolate the first of those possibilities.

2.6. Timing and Agent-Level Credit

A terminal welfare score can describe the desired outcome clearly and still be hard for the learner to use. Delay is the problem. Pignatelli et al. [26] review temporal credit assignment as the task of linking a later outcome to the earlier actions that helped produce it.

2. Background

RUDDER, proposed by Arjona-Medina et al. [27], redistributes delayed return toward earlier state-action pairs that are more informative about the final result. Zhai et al. [28] study intermediate rewards from another angle and report benefits in long-horizon goal-reaching problems. Earlier information can help—and it can also mislead.

Ng, Harada, and Russell [29] explain why. Potential-based shaping preserves the optimal policy under defined conditions, whereas arbitrary shaping can change what the agent is actually optimising. A dense fairness signal therefore needs more than early timing.

Multi-agent learning adds a second problem: the signal may arrive at the right time but reach the wrong agent. Tumer and Agogino [30] use difference rewards to compare system performance with a counterfactual in which one agent’s contribution is removed. Foerster et al. [31] use a central critic in COMA to compare one agent’s selected action with alternatives while holding the other agents’ actions fixed.

Mashayekhi et al. [32] study prosocial norm emergence through accumulated costs in local interactions. Their setting differs from highway merging, but the idea transfers: persistent disadvantage can be more informative than a one-step snapshot when deciding whether a prosocial response is needed.

The two credit problems should stay separate. DWS later changes **when** welfare feedback is delivered. EXP05 also changes **which local disadvantage** is represented and how the ego action is treated.

2.7. Technical Foundations

2.7.1. Highway-Env

Leurent [33] introduced `highway-env` for autonomous-driving decision research. It provides configurable road networks, vehicle dynamics, kinematic observations, and discrete or continuous control; road layout, traffic population, observation fields, actions,

and rewards can all be altered through the environment interface.

That flexibility matters here. The experiments use `highway-env==1.12.0`, but not the standard task unchanged. [chapter 3](#) defines the four-vehicle merge built on top of it.

2.7.2. Deep Q-Networks and Double DQN

Mnih et al. [\[34\]](#) introduced the Deep Q-Network, using a neural network to approximate action values instead of storing a separate table entry for every state–action pair. Their Atari experiments showed that value-based reinforcement learning could generalise across high-dimensional observations.

Standard DQN can overestimate action values. Van Hasselt, Guez, and Silver [\[35\]](#) address this with Double DQN by separating action selection from action evaluation in the target update. The controller in this thesis follows that variant.

2.7.3. Parameter Sharing

Gupta, Egorov, and Kochenderfer [\[36\]](#) study cooperative deep RL under partial observability and include decentralised parameter sharing among the methods they evaluate. Several agents reuse the same network weights, but each agent feeds in its own observation—one shared policy, four distinct decisions.

Chen et al. [\[1\]](#) apply the same idea to highway on-ramp merging, where parameter sharing supports coordination across multiple controlled vehicles without training one independent network per vehicle. That is the architecture adopted here.

2.7.4. Curriculum Learning

Curriculum learning changes the order in which difficulty appears. Chen et al. [\[1\]](#) use it in cooperative highway merging so that policies see harder traffic settings after simpler ones.

2. Background

The curriculum in this thesis is deliberately narrower: only scenario diversity changes. Training starts with one merge scenario, expands to 4, then 16, and finally 64. Fairness is not introduced during this stage; the task policy is learned first. [chapter 3](#) gives the exact step budget for each stage.

2.8. Research Gap

Highway coordination is already learnable [\[1\]](#), [\[4\]](#), and fair sequential decision-making is well established through Generalized Gini, Maximin, inequity aversion, specialised architectures, or social reward design [\[14\]](#), [\[16\]](#), [\[19\]](#), [\[20\]](#), [\[25\]](#). The missing comparison is narrower than either literature alone suggests.

Most successful fair-MARL systems change several components at once. FEN changes reward and hierarchy [\[19\]](#); Zimmer et al. [\[20\]](#) introduce a dedicated two-part architecture; Toghi et al. [\[24\]](#) redesign distributed rewards; Hughes et al. [\[25\]](#) add accumulated inequity. A favourable final distribution therefore does not isolate the contribution of the social-welfare aggregation rule.

EXP02 changes that rule and little else. Mean, GGI, and Rawlsian Maximin continue from matched starting policies while the highway task, DQN architecture, local observation, task reward, and training budget stay fixed. WSC and DWS follow as diagnostics rather than as new formal research questions.

EXP05 is intentionally less isolated. It combines accumulated burden, visible-neighbour comparison, and action-dependent feedback in one late follow-up. The package can be evaluated as a whole; its individual components cannot be credited separately.

3. System Design and Core Method

3.1. Study Design

The formal experiment uses four learned vehicles in one controlled merge: Fast and Slow on the ramp, and Fast and Slow on the mainline. Every condition keeps the same simulator, discrete action interface, and parameter-shared controller. What changes is the fairness mechanism.

The sequence is staged. EXP01 measures the task-only welfare distribution. EXP02 changes the terminal social-welfare objective while leaving the rest of the setup matched. WSC then adds welfare information to the observation, DWS moves welfare feedback into the episode, and EXP05 combines a burden representation with local action-dependent feedback.

Earlier development tasks are not part of the formal comparison. [chapter D](#) summarises them separately.

3.2. Highway-Merging Environment

3.2.1. Road Geometry and Episode Termination

The study instantiates a four-vehicle merge in `highway-env==1.12.0` [33]. Decisions are made every 0.2 s. An episode lasts at most 200 policy steps.

The route is 500 m long: 220 m before convergence, an 80 m converging section, an

3. System Design and Core Method

80 m parallel merge, and 120 m after the merge. Vehicles count as complete at 470 m if they have not collided. The episode stops when all four complete, when a controlled vehicle collides, or when the step limit is reached. Collision and timeout are failures.

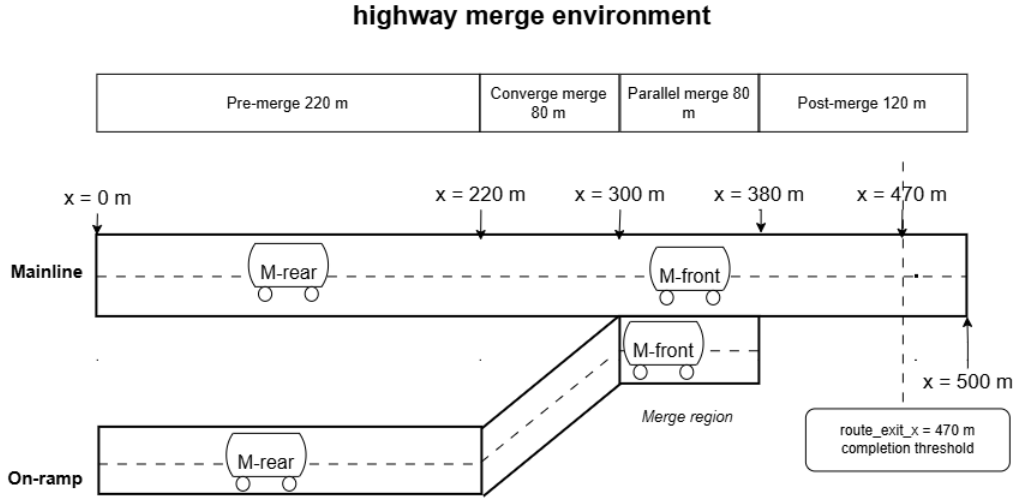


Figure 3.1.: Highway-merging road geometry used in the study.

3.2.2. Vehicle Heterogeneity

Each road carries one Slow and one Fast controlled vehicle. Slow vehicles have an 18 m/s target; Fast vehicles have a 22 m/s target. Those values define the mobility requirements used by the welfare measure.

The learned controller does not change them. It changes driving behaviour around fixed requirements.

3.2.3. Matched Time-to-Conflict Initialisation

Starting distance is not held equal across the two speed classes. Instead, vehicles are placed using approximate time-to-conflict (TTC), which removes the trivial advantage a faster target speed would create if every vehicle started the same distance from the merge.

Two nominal arrival waves create the base interaction. Small random jitters change the exact ordering from episode to episode, and invalid same-lane configurations are rejected when vehicles would start too close together. [sections B.1](#) and [B.2](#) give the exact generation rules.

3.3. Policy Architecture and Training

3.3.1. Parameter-Shared DQN

The controller implements the parameter-shared Double DQN introduced in [section 2.7](#). One network is used by all four vehicles. Each vehicle passes in its own local observation, so the shared weights do not force the vehicles to take the same action.

Two hidden layers are used, each with 64 units. Evaluation is deterministic: exploration is disabled and the action with the highest predicted Q-value is selected. Adam [\[37\]](#) optimises the network. Remaining optimisation settings are listed in [section B.3](#).

3.3.2. Curriculum Task Learning

Fairness training does not begin from a randomly initialised policy. The shared controller first learns the merge under the Baseline task reward.

Training moves through four scenario sets. C1 contains one scenario and lasts 400,000 steps. C4 adds three more scenarios for 300,000 further steps. C16 then runs for 250,000 steps, followed by C64 for another 250,000. The result is the common 1.2M-step task

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checkpoint.

Within a seed, that exact checkpoint is copied into the Baseline, Mean, GGI, Rawlsian Maximin, WSC, and DWS continuations. EXP05 starts from a later 3.0M Baseline checkpoint. The table in [section B.3](#) records the same schedule in compact form.

3.4. Local Observation

3.4.1. Original Observation

Each vehicle receives an 18-dimensional local observation containing its own road role, speed, target, recent control state, merge position, and action history, plus relative information for up to three neighbours within 50 m. It contains no global welfare distribution or accumulated welfare.

3.4.2. Welfare History and WSC

Target attainment is $u_{i,t} = \text{clip}\left(\frac{v_{i,t}}{v_i^{\text{target}}}, 0, 1\right)$. The welfare-history variable $M_i(t) = \frac{1}{n_i(t)} \sum_{k=1}^{n_i(t)} u_{i,k}$ is the mean target attainment experienced by vehicle i up to decision time t , with $M_i(t) = 1$ before the first active sample. It therefore represents an episode prefix rather than instantaneous welfare. WSC adds the ego and three visible-neighbour welfare-history values, increasing the observation from 18 to 22 dimensions. The full layout is in [section B.3](#).

3.5. Action and Longitudinal Control

The policy chooses ACCELERATE, HOLD, or BRAKE, which change a desired-speed setpoint used by the simulator’s low-level controller. Target speed remains fixed and defines the welfare requirement; desired speed is the controller setpoint; actual speed is the realised

vehicle speed. HOLD keeps the setpoint unchanged and therefore does not imply zero physical acceleration. Controller parameters are reported in [section B.3](#).

3.6. Individual Welfare and Burden

3.6.1. Final Episode Utility

Let $S_i = 1$ when vehicle i completes safely and $S_i = 0$ after collision, timeout, or incompletion. Final utility is $U_i = S_i \frac{\sum_t I_{i,t}^{\text{active}} u_{i,t}}{\sum_t I_{i,t}^{\text{active}}}$. Thus U_i combines target attainment with task success: a failed vehicle receives $U_i = 0$. Welfare history $M_i(t)$ is available during the episode; U_i is the episode-end individual outcome.

3.6.2. Requirement-Relative Mobility Burden

Accumulated mobility loss is measured as $C_i = \Delta t \sum_t I_{i,t}^{\text{active}} (1 - u_{i,t})$. Burden increases with the duration and size of target shortfall. It is not equivalent to counting BRAKE actions, because braking can occur while a vehicle remains above target.

3.6.3. Worst-Off Utility and Utility Gini

The two primary fairness outcomes summarise the four final utilities $\mathbf{U} = (U_1, U_2, U_3, U_4)$. They are evaluation statistics, not training rewards.

Worst-off utility is $U_{\min} = \min_i U_i$. It records the lowest individual outcome in the episode. Because a failed vehicle has $U_i = 0$, any collision, timeout, or incompletion sets $U_{\min} = 0$.

Utility Gini is $G_U = \sum_{i=1}^4 \sum_{j=1}^4 |U_i - U_j| / (2 \cdot 4 \cdot \sum_{k=1}^4 U_k)$. It equals 0 when the four utilities are equal and increases as they spread. If all four utilities are 0, the denominator is zero and G_U is undefined; those episodes are omitted from the seed-level Gini mean.

The two numbers can move separately: a lower G_U does not imply a higher $U_{\min}$.

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Utility Gini is also not the GGI training objective in [section 3.7](#). GGI is a weighted welfare level; G_U is an inequality index on the same four utilities.

3.7. Reward Design

3.7.1. Baseline Task Reward

The Baseline reward encourages route progress and completion and penalises collision, hard braking, and timeout. Exact coefficients are in [section B.3](#). This task reward remains present in every welfare condition.

3.7.2. Social-Welfare Evaluation

For final utilities $\mathbf{U} = (U_1, U_2, U_3, U_4)$, the three welfare objectives are

$$\begin{aligned} W_{\text{Mean}}(\mathbf{U}) &= \frac{1}{4} \sum_{i=1}^4 U_i, \\ W_{\text{GGI}}(\mathbf{U}) &= 0.4U_{(1)} + 0.3U_{(2)} + 0.2U_{(3)} + 0.1U_{(4)}, \\ W_{\text{Maximin}}(\mathbf{U}) &= \min_i U_i, \end{aligned}$$

where $U_{(1)} \leq \dots \leq U_{(4)}$.

The GGI weights were fixed before formal training.

3.7.3. Combining Task and Welfare Rewards

Mean, GGI, and Rawlsian Maximin add the terminal term $R_c^W = 0.5[W_c(\mathbf{U}) - 1]$ to the Baseline task reward at episode termination. The comparison therefore changes how the final welfare distribution is valued without replacing the traffic task reward. WSC changes observation information instead, while DWS and EXP05 change intermediate welfare feedback.

3.8. Evaluation and Statistical Principles

A **training seed** is one independent training run, a **checkpoint** is a saved network during training, and the **endpoint** is the checkpoint used for the main result. Formal testing uses H1, a fixed set of 256 merge scenarios not used for training. Episode outcomes are averaged within each seed, and uncertainty is calculated across the 12 independently trained policies.

Matched comparisons subtract the Baseline result from the corresponding welfare-policy result within the same seed. Positive $\Delta U_{\min}$ and negative ΔG_U are favourable (section 3.6.3). Confidence intervals use 10,000 bootstrap resamples of the 12 seed-level differences; intervals crossing zero are treated as uncertain in direction.

The main text treats saved checkpoints as the policy objects of interest. A historical late-checkpoint averaging audit is retained in section C.1 for transparency, but it is not introduced as a separate experimental condition. EXP05 uses the single pre-specified 3.8M network.

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4.1. Development Strategy

EXP01 and EXP02 address the two formal research questions. WSC and DWS were added as diagnostic experiments after the objective comparison, and EXP05 was added as a final mechanism-guided follow-up. The sequence is problem-driven: each later experiment changes a component that the previous experiment leaves unresolved.

4.2. EXP01 — Baseline Welfare

EXP01 asks whether task-only coordination produces an uneven distribution of mobility welfare and, if so, what that disadvantage looks like.

The Baseline learning curve is shown from the start of curriculum training so that the final welfare analysis can be read in the context of the complete task-learning process.

4.2.1. Finding 1: Task Success Does Not Eliminate Welfare Inequality

Before comparing fairness objectives, the Baseline must perform the coordination task well enough for its welfare distribution to be meaningful. The formal analysis uses the 12 Baseline policies evaluated on the fixed H1 test set.

Across seeds, completion is 0.884, collision is 0.116, and no timeout is observed. Mean individual utility is 0.941. Mean worst-off utility is lower at 0.882, while Utility Gini is

4. Experimental Development

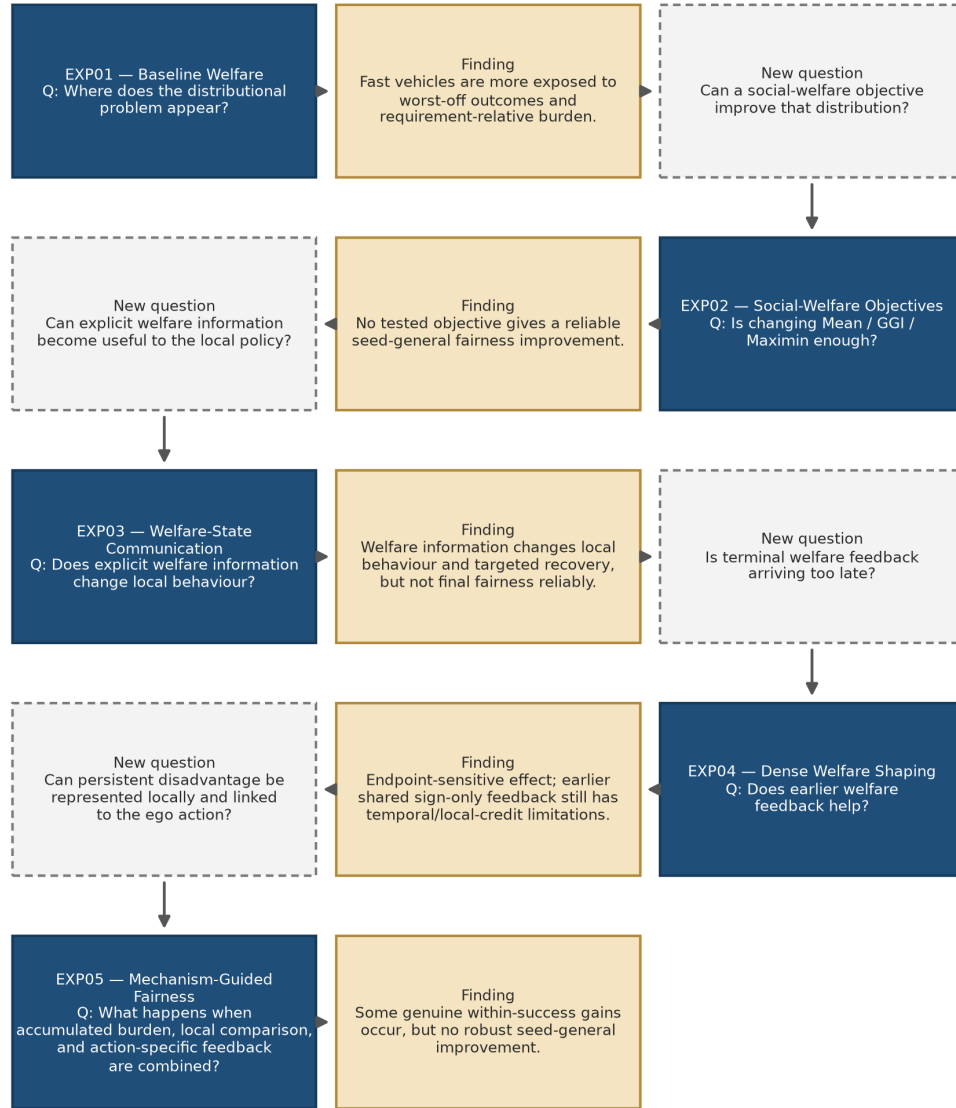


Figure 4.1.: Experimental sequence from Baseline analysis to the objective comparison, diagnostic experiments, and final follow-up.

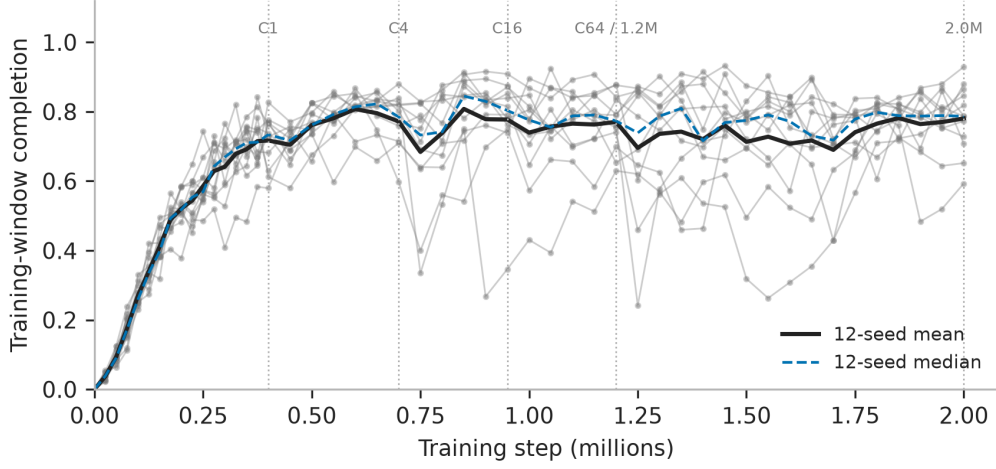


Figure 4.2.: Baseline training-window completion from step 0. Vertical markers indicate curriculum and continuation boundaries.

0.058 with a 95% seed-level interval of $[0.025, 0.096]$.

The Baseline therefore usually completes the merge, but successful task learning does not remove variation in the welfare experienced by the four vehicles.

4.2.2. Finding 2: Higher Mobility Requirements Are More Exposed to the Welfare Lower Tail

Average welfare is similar for Fast and Slow vehicles: the Fast-minus-Slow difference is only -0.006 , and its confidence interval crosses zero. The lower tail shows a clearer asymmetry. Fast vehicles account for 69.8% of identifiable worst-off outcomes, with a 95% interval of $[55.5\%, 83.3\%]$.

Fast vehicles therefore do not have uniformly lower utility. Instead, they are more frequently represented among the worst outcomes.

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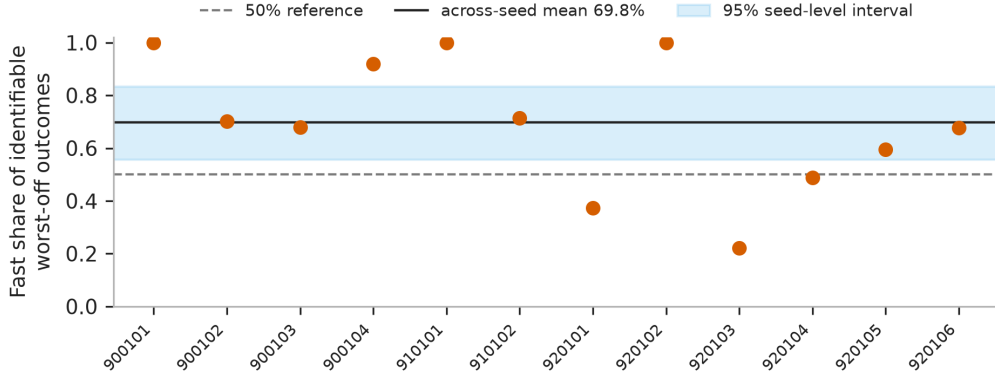


Figure 4.3.: Fast-vehicle share of identifiable worst-off outcomes across the 12 Baseline training seeds. The dashed line marks 50%.

4.2.3. Finding 3: Braking Behaviour Does Not Identify Welfare Disadvantage

A natural interpretation is that the vehicle that brakes more often also carries the greater coordination burden. The Baseline data show that this is not the case.

Slow vehicles choose **BRAKE** on 8.94% of relevant steps, compared with 5.52% for Fast vehicles. However, Slow vehicles are still fully meeting their target requirement during 96.7% of those braking steps, compared with 83.7% for Fast vehicles.

The welfare measures reverse the behavioural pattern. Slow vehicles spend 1.37% of active time below target, compared with 3.26% for Fast vehicles. When below target, mean relative shortfall is 0.087 for Slow vehicles and 0.135 for Fast vehicles. The Fast-minus-Slow mobility-burden difference is +0.042, with a 95% interval of [+0.004, +0.093].

Slow vehicles therefore make the larger visible concession, whereas Fast vehicles experience the larger requirement-relative mobility loss. Braking frequency and welfare burden are not interchangeable measures of disadvantage. Detailed physical-concession measures are retained in [section C.5](#).

The same direction remains visible across merge phases and within successful episodes;

4.3. EXP02 — Social-Welfare Objectives

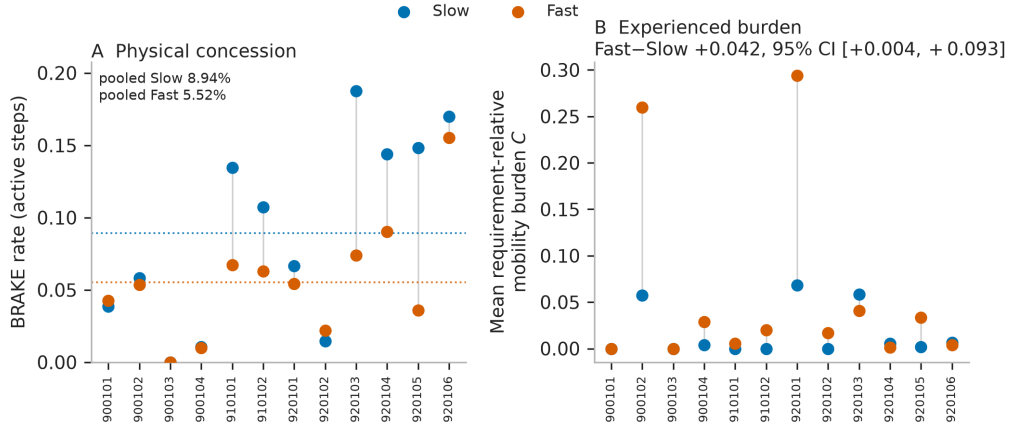


Figure 4.4.: Seed-level physical-concession and requirement-relative mobility-burden summaries for Slow and Fast vehicles.

supporting checks are reported in [section C.5](#).

4.2.4. Overall Finding of EXP01

EXP01 supports RQ1. Fast vehicles are disproportionately represented in the welfare lower tail, while visible concession points to a different group from requirement-relative burden. EXP02 next tests whether changing the social-welfare objective can improve that distribution.

4.3. EXP02 — Social-Welfare Objectives

4.3.1. Experimental Question and Matched Design

Each of the 12 training seeds produces one common Baseline checkpoint at 1.2M steps. That checkpoint is copied into four branches. One branch continues with the task reward only; the other three add the Mean, GGI, or Rawlsian Maximin terminal welfare reward defined in [section 3.7](#). Every branch then trains for another 800,000 steps.

4. Experimental Development

All four conditions therefore share the same task-learning history within a seed up to the 1.2M branch point. The learning curve should show that common history once and separate the four conditions only after they actually diverge.

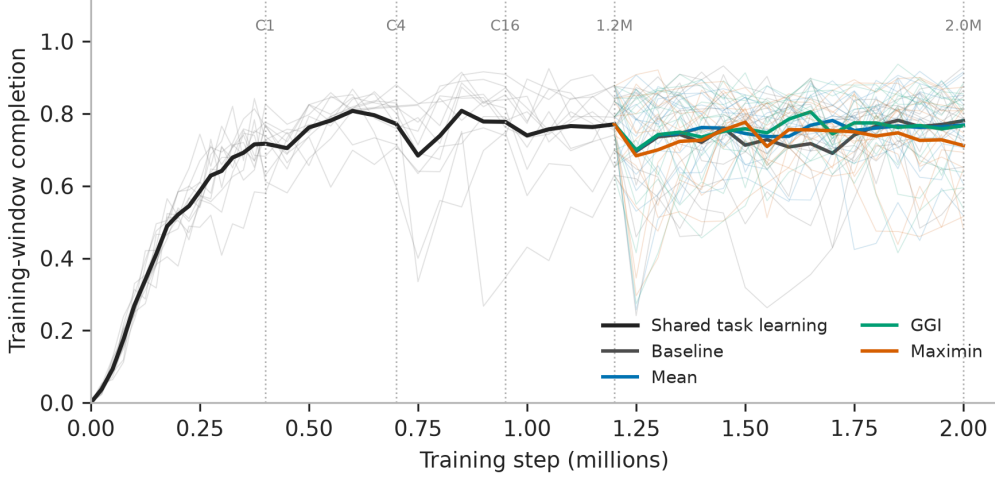


Figure 4.5.: Training-window completion from step 0 for the shared task-learning history and the Baseline, Mean, GGI, and Rawlsian Maximin continuations. Vertical markers indicate curriculum boundaries and the 1.2M branch point.

4.3.2. Task Performance as Context

The four conditions remain capable of performing the merge. Completion is 0.884 for Baseline, 0.903 for Mean, 0.895 for GGI, and 0.872 for Rawlsian Maximin. Rawlsian Maximin also records a timeout rate of 0.020, the largest of the four conditions. [section C.6](#) reports the complete task-outcome table.

These task outcomes are context rather than fairness results. Final utility is failure-aware, so a collision, timeout, or incompleteness can drive a vehicle’s final utility to zero. The fairness comparison therefore focuses on paired changes in worst-off utility and Utility Gini while reporting task outcomes alongside them.

4.3.3. Finding 1: Changing the Welfare Objective Alone Does Not Reliably Improve the Distribution

The mean paired change in worst-off utility is small for all three welfare objectives. Mean gives $\Delta U_{\min} = +0.020$, GGI gives $+0.013$, and Rawlsian Maximin gives -0.008 . Their 95% intervals are $[-0.129, +0.139]$, $[-0.054, +0.093]$, and $[-0.079, +0.062]$, respectively.

The seed counts show the same variability. Mean improves $U_{\min}$ in 8 of 12 matched runs, while GGI and Rawlsian Maximin each improve 6. Utility Gini also changes only slightly: Mean reduces it by about 0.009 on average and GGI by about 0.008, while Rawlsian Maximin is nearly unchanged. All three Utility Gini intervals include zero.

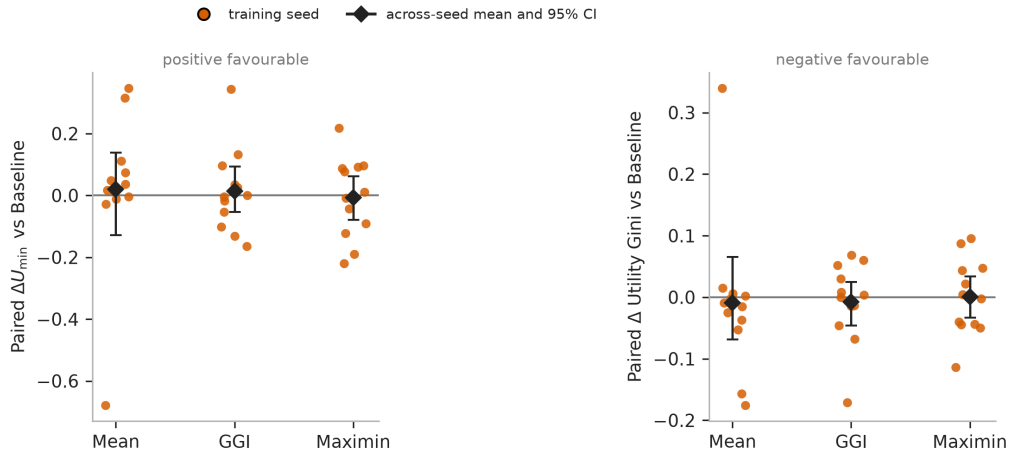


Figure 4.6.: Seed-paired changes in worst-off utility and Utility Gini relative to the matched Baseline.

The objective change alone therefore does not produce a seed-general fairness improvement in this setting. Mean and GGI move in a favourable direction on average, but the direction is not reproduced reliably across the 12 independent training runs.

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4.3.4. Finding 2: Rawlsian Maximin Provides Little Additional Separation Among Successful Episodes

Among successful episodes, median $U_{\min}$ is 1.0 under both Baseline and Rawlsian Maximin. Baseline has $U_{\min} > 0.98$ in 93.2% of successful episodes, compared with 89.6% under Rawlsian Maximin. Restricting the analysis to successful episodes gives a paired change of approximately $\Delta U_{\min} = -0.010$, with a 95% interval of $[-0.028, +0.002]$.

A check on the terminal reward supports the same interpretation. About 90.1% of the variation in the Rawlsian Maximin bonus is accounted for by whether an episode ends in success, collision, or timeout. Once the merge has succeeded, the bonus has relatively little remaining variation with which to distinguish one successful welfare distribution from another.

This does not establish a zero within-success effect. It shows that the experiment provides little evidence that the terminal Rawlsian Maximin bonus improves welfare redistribution once successful coordination has already been achieved.

4.3.5. What EXP02 Leaves Open

EXP02 answers RQ2 through a controlled social-welfare objective comparison: changing only the terminal aggregation rule does not reliably improve the final welfare distribution.

A practical question remains. Under the Original observation, a vehicle can see local traffic state but cannot see the welfare already experienced by itself or its neighbours. EXP03 tests whether making that history observable changes local behaviour.

4.4. EXP03 — Welfare-State Communication

EXP03 tests one possible missing link from EXP02: whether the local policy can use explicit information about welfare already experienced during the episode.

4.4.1. Experimental Design

WSC expands the local observation from 18 to 22 dimensions by adding welfare history $M_i(t)$ for the ego vehicle and the three visible-neighbour slots. The traffic state, reward definition, and 50 m visible range remain unchanged. Baseline, Mean, GGI, and Rawlsian Maximin are continued from the same 1.2M task checkpoint under both Original and WSC observations.

The WSC network expansion copies the original 18 first-layer columns and zero-initialises the four new welfare-history columns. Before continuation, the transformed network reproduces the source Q-values to within 4.77×10^{-7} and gives 100% selected-action agreement on 2,208 matched observations. This check isolates the observation change from an unintended initial policy change.

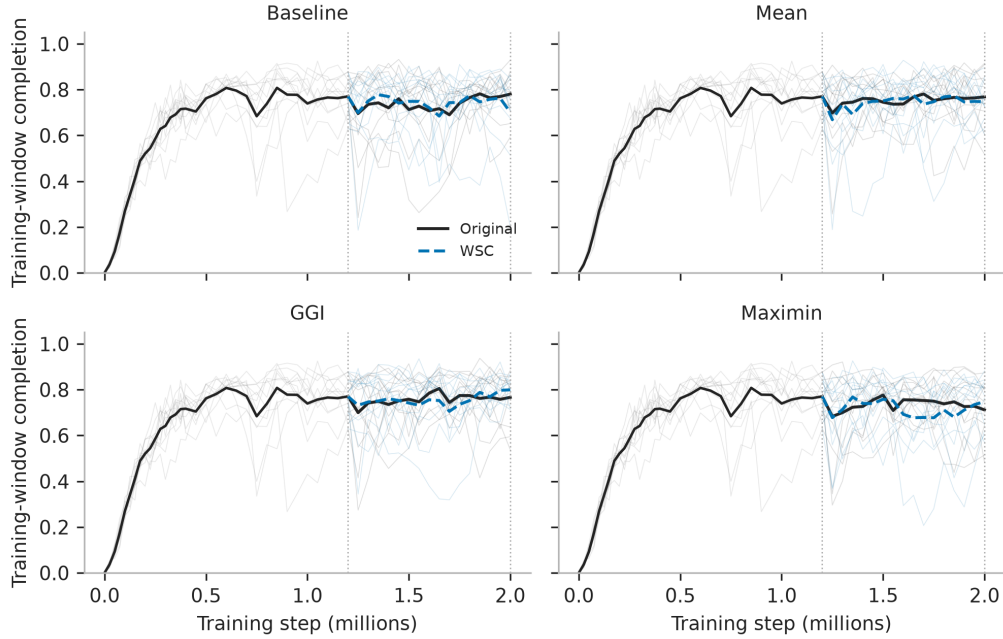


Figure 4.7.: Training-window completion from step 0 for the matched Original and WSC continuations. Vertical markers indicate the shared training history and WSC branch point.

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4.4.2. Finding 1: WSC Does Not Reliably Improve Final Fairness

The interaction analysis asks whether adding WSC changes the welfare effect of the Mean, GGI, or Rawlsian Maximin objective relative to the corresponding Original-observation condition. The most favourable average interaction is observed for GGI: the extra WSC effect is $+0.074$ on $U_{\min}$ and -0.038 on Utility Gini. Mean and Rawlsian Maximin do not show the same favourable average pattern.

The uncertainty remains substantial. Every 95% interaction interval includes zero, so the data do not provide reliable evidence that WSC improves final fairness across training seeds. Full interaction estimates are reported in [section C.7](#).

4.4.3. Finding 2: The Trained Policies Use the Welfare-History Inputs

The weak endpoint interaction does not mean that the added information is ignored. This can be tested by changing the welfare-history inputs of an already trained WSC policy while leaving the network weights and physical traffic state unchanged.

In the live masking test, the four welfare-history inputs are replaced by neutral values without retraining. Mean $U_{\min}$ then falls by 0.0161, Utility Gini rises by 0.0077, completion falls by 1.99 percentage points, and collision rises by 1.49 percentage points.

The changes are modest, but the intervention is direct: the network weights and physical observations are unchanged. The trained policy therefore depends on the welfare-history channels during interaction. Supporting fixed-state perturbations are retained in [section C.7](#).

4.4.4. Finding 3: WSC Supports Local Welfare Recovery After Targeted Yielding

Welfare history also contains information about who may remain disadvantaged later in the episode. At the episode midpoint, the vehicle with the lowest welfare history is the

final worst-off vehicle in 87.96% of episodes.

The clearest behavioural result appears in targeted yielding. The audit identifies 24,378 events in which a vehicle yields while a visible neighbour is already worse off according to welfare history. The audit records a targeted-yielding event when the ego vehicle takes BRAKE while a visible neighbour’s running welfare M is at least 0.02 below the ego vehicle. Recovery at horizon $h \in \{10, 25, 50\}$ is the event-level fraction of those cases in which the disadvantaged neighbour’s M at step $t + h$ is strictly larger than its M at the yielding step. It is therefore a percentage of events that recover, not a percentage increase in the neighbour’s utility. For example, 65.6% recovery at +10 means that 65.6% of the validated WSC yielding events satisfy $M_{\text{neighbour}}(t + 10) > M_{\text{neighbour}}(t)$.

Within ten steps, recovery is 23.1% under Original observations and 65.6% under WSC; the difference remains large at 25 and 50 steps. This is more informative than a general change in braking frequency because it links an observable disadvantage, a local concession, and a subsequent recovery event. The 24,378 audited events are used descriptively and are not treated as independent training replications.

4.4.5. Overall Finding of EXP03 and Motivation for DWS

EXP03 provides a positive mechanism result even though its endpoint fairness effect is uncertain. The trained policies use welfare history, and targeted yielding under WSC is followed by substantially more local welfare-gap recovery. Making disadvantage observable can therefore change behaviour in a way that is locally consistent with recovery.

WSC does not solve the learning-timing problem. It changes what the policy can observe, but the main social-welfare objective still evaluates the completed episode. A vehicle can see that a neighbour is disadvantaged without receiving an immediate learning signal that connects its current action to the eventual group outcome. EXP04 therefore tests whether moving welfare-related feedback into the episode reduces this temporal

4. Experimental Development

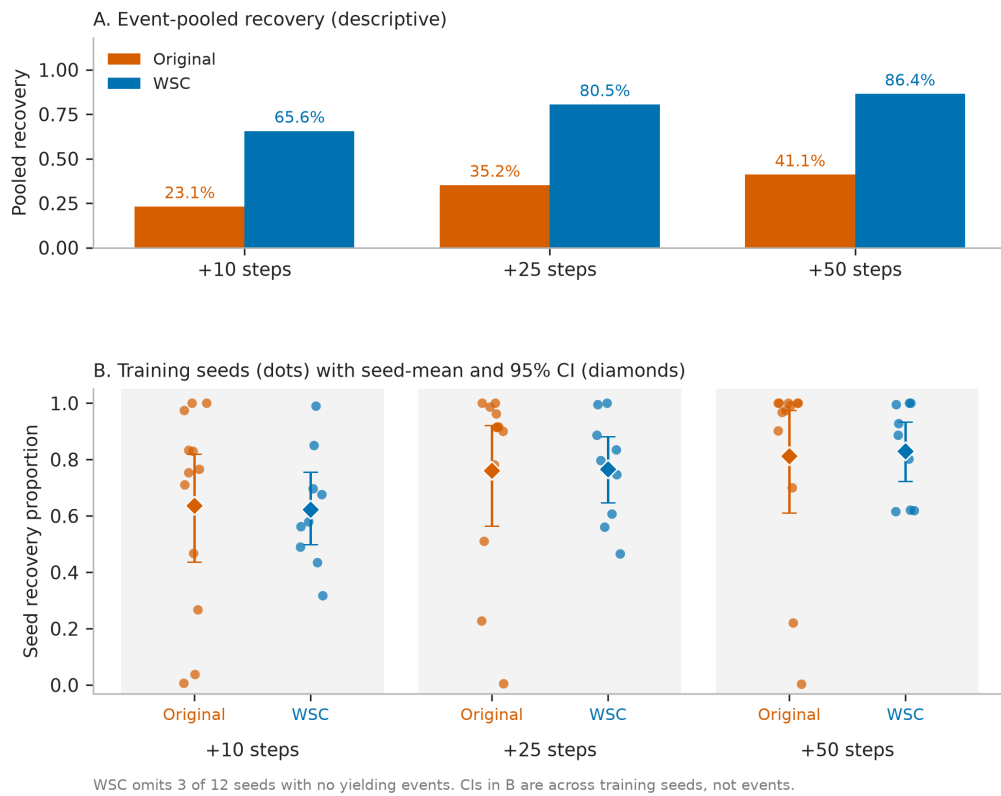


Figure 4.8.: Panel A reports welfare-gap recovery after targeted yielding under Original observations. Panel B reports the corresponding recovery under WSC.

credit gap.

4.5. EXP04 — Dense Welfare Shaping

EXP04 tests whether earlier welfare-related feedback can make the fairness objective easier to learn from than a terminal signal alone.

4.5.1. DWS Design

Welfare history $M_i(t)$ summarises how well vehicle i has met its target up to time t , but unlike final utility U_i , it does not yet include whether the episode will later succeed or fail.

DWS forms a welfare-history floor, $\Phi_t = \min_i M_i(t)$, and adds a small shared signal according to whether that floor rises or falls:

$$F_t = \begin{cases} +0.0005, & \Phi_{t+1} - \Phi_t > 10^{-6}, \\ -0.0005, & \Phi_{t+1} - \Phi_t < -10^{-6}, \\ 0, & \text{otherwise.} \end{cases}$$

The terminal Rawlsian Maximin reward remains in place. The same F_t is sent to all active agents, so DWS makes welfare feedback earlier and denser but does not identify which particular agent should respond.

4.5.2. Finding 1: DWS Shows a Favourable Average Shift at the Single 2.0M Endpoint, but Not a Consistent Seed-Level Direction

Evaluating the actual 2.0M Original+DWS network gives an average paired change of approximately $\Delta U_{\min} = +0.0663$ and $\Delta G_U = -0.0243$. Both averages move in the favourable direction, but six seeds move in each direction, so the improvement is not consistent across independently trained policies.

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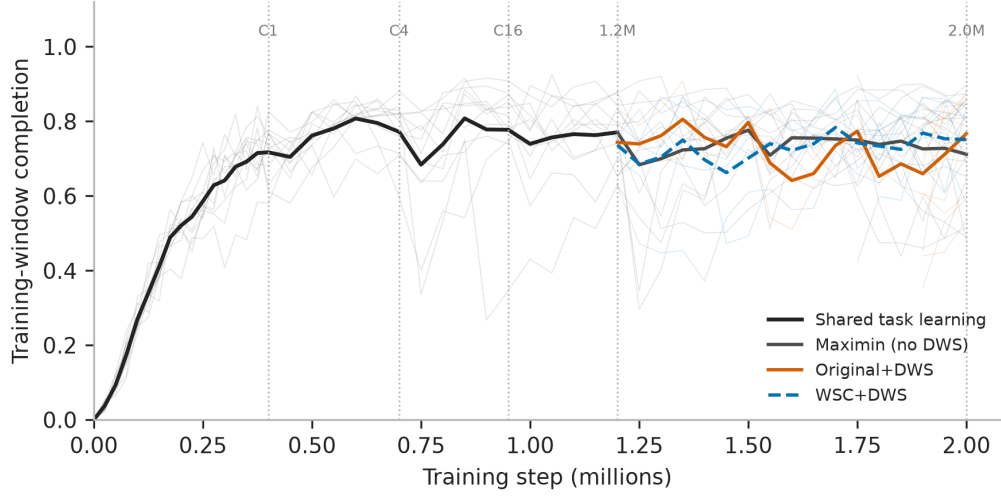


Figure 4.9.: Training-window completion from step 0 for the DWS conditions and their matched pre-intervention training history. Vertical markers indicate the DWS branch point and final endpoint.

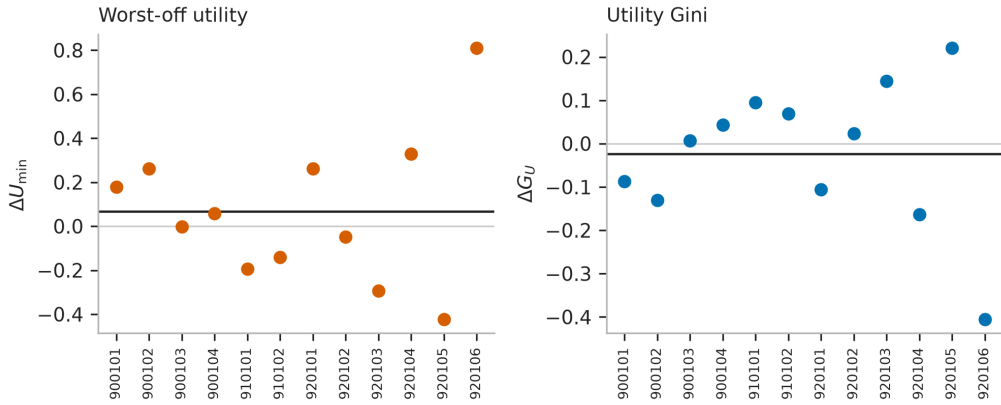


Figure 4.10.: Seed-paired Original+DWS changes in worst-off utility and Utility Gini at the single 2.0M checkpoint.

The main conclusion is therefore limited: DWS can move the average endpoint in a favourable direction, but the available seeds do not establish a stable effect. The historical checkpoint-representation audit is retained in [section C.1](#) rather than used as the main experimental narrative.

4.5.3. Finding 2: Earlier Feedback Reveals Temporal and Local Credit Gaps

The DWS logs help explain why a dense shared signal can remain difficult to use. First, the welfare-history floor and final worst-off utility are related but not identical. An episode-level summary of the welfare-history floor correlates with final $U_{\min}$ at about $r = 0.727$, yet about 8.3% of episodes receive more positive than negative DWS feedback and still finish with $U_{\min} < 0.5$. Final utility also depends on whether the episode ultimately succeeds.

Second, the effect of a local action can take time to appear. After a negative DWS step associated with braking, recovery is only 0.1% under Original and 3.1% under WSC after five steps. By 50 steps, those values rise to 57.4% and 65.6%. A short-horizon change in the welfare-history floor can therefore look unfavourable even when the later trajectory recovers.

Third, the same DWS signal is sent to every active agent. In the audit, 41.7% of recipients under Original+DWS and 24.4% under WSC+DWS are not locally involved in the event that produced the signal. Earlier feedback therefore does not by itself identify local responsibility. Supporting diagnostics are reported in [section C.8](#).

4.5.4. Overall Finding of EXP04

EXP04 shows that earlier welfare feedback is not sufficient when the signal can disagree with the final outcome, precede a longer-term recovery, or be shared with agents that

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were not locally involved. This shifts the remaining design problem from timing alone to the representation of persistent disadvantage and local action credit.

4.6. EXP05 — Mechanism-Guided Fairness

4.6.1. Motivation from Earlier Experiments

EXP05 combines three mechanism lessons from the previous experiments. EXP01 showed that visible braking and experienced mobility burden can identify different disadvantaged groups, motivating a persistent requirement-relative burden representation rather than an action count. EXP03 showed that policies can use welfare history and that local yielding can be followed by welfare recovery, motivating explicit information about disadvantaged neighbours. EXP04 showed that an earlier shared signal still leaves responsibility diffuse, motivating feedback that depends on the ego vehicle’s own action.

The resulting follow-up therefore combines accumulated burden, visible-neighbour comparison, and action-dependent local feedback while retaining terminal Rawlsian Maximin. The experiment evaluates the package as a whole; because the three changes are introduced together, it cannot identify which component is independently responsible for any observed change.

4.6.2. Experimental Design and Training Base

EXP05 starts from the 3.0M Baseline policies and continues training for 800,000 steps to the pre-specified 3.8M endpoint. The continuation retains terminal Rawlsian Maximin and adds the three mechanism-guided changes described in [section 4.6.3](#). The 3.0M starting point was fixed before the final fairness result because the development seeds had broadly consolidated task performance by that stage, whereas extending task-only training to 3.5M was mixed ([section C.2](#)).

There is no parallel 3.0M-to-3.8M control continuation without the new mechanism. EXP05 is therefore a matched pre-to-post follow-up, not a causal treatment-control comparison. Detailed continuation hyperparameters are reported in [section C.9](#).

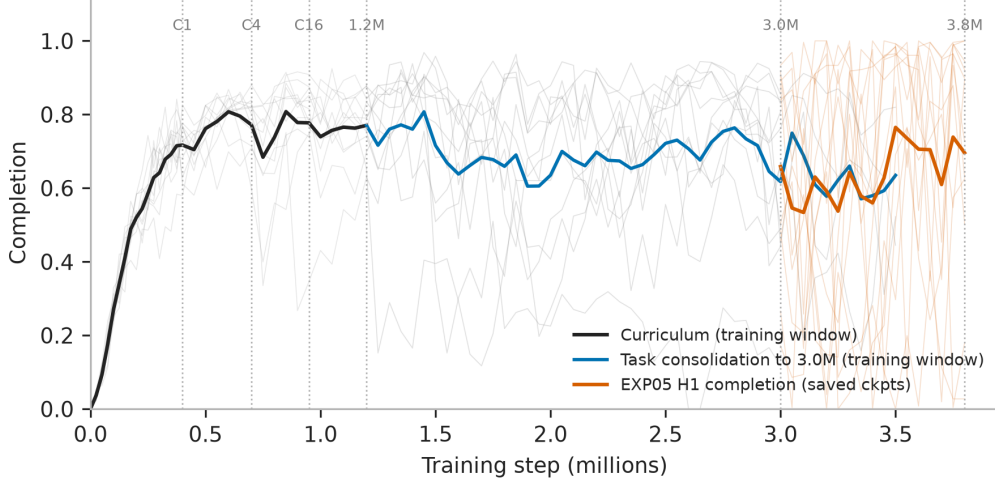


Figure 4.11.: EXP05 training trajectory from step 0 through the 3.0M training base and the 3.8M endpoint. Vertical markers indicate the intervention start and final endpoint.

4.6.3. Mechanism Implementation

EXP05 implements the earlier mechanism lessons in three steps:

1. **Accumulated burden representation.** For each active vehicle, $u_i(t) = \text{clip}\left(\frac{v_i(t)}{v_i^{\text{target}}}, 0, 1\right)$ and accumulated mobility burden before decision t is

$$C_i(t) = \sum_{\tau < t} I_{\text{active},i}(\tau) \Delta t [1 - u_i(\tau)].$$

Normalised burden is $b_i(t) = \text{clip}\left(\frac{C_i(t)}{C_{\text{max}}}, 0, 1\right)$. EXP05 replaces the WSC welfare-history channels with ego and visible-neighbour burden.

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2. **Visible-neighbour burden comparison.** The local gap $g_i(t) = \max_{j \in \mathcal{N}_i(t)} [b_j(t) - b_i(t)]$ records whether a visible neighbour has accumulated more burden than the ego vehicle. Small differences are ignored; above threshold c , severity grows as $\text{severity}_i(t) = \text{clip}\left(\frac{g_i(t) - c}{g_{95} - c}, 0, 1\right)$. The development seeds fixed $c \approx 0.0078$ and $g_{95} \approx 0.0280$; exact values are in [section C.9](#).
3. **Action-dependent local feedback.** Ego actions receive weights $w(\text{BRAKE}) = 0$, $w(\text{HOLD}) = 0.5$, and $w(\text{ACCELERATE}) = 1$, giving $r_i^{\text{local}} = -0.0015 \text{severity}_i(t) w(a_i)$. The extra penalty is therefore strongest when the ego vehicle continues to accelerate while a visible neighbour already carries more burden; braking is not positively rewarded. The terminal Rawlsian Maximin term remains $R_{\text{Maximin}}^W = 0.5[\min_i U_i - 1]$.

4.6.4. Evaluation

For each seed, the paired change is computed within the same training run:

$$\Delta U_{\min} = U_{\min}^{3.8M, \text{EXP05}} - U_{\min}^{3.0M, \text{Baseline}}.$$

A positive value means that the 3.8M EXP05 policy has higher worst-off welfare than its own 3.0M starting policy. Averaging these 12 within-seed changes reduces variation caused by different starting policies, but the absence of a parallel continuation control means that the paired change should not be interpreted as the causal effect of EXP05.

Overall $U_{\min}$ is the primary outcome. Success-only $U_{\min}$ is reported to distinguish redistribution within already-successful merges from changes driven mainly by the mix of successful and failed episodes. Supporting outcomes—Utility Gini, mean utility, completion, collision, timeout, and burden measures—are used to interpret the primary result rather than as independent claims of success.

4.6.5. Finding 1: EXP05 Produces a Favourable Average Shift in Worst-Off Welfare

At 3.8M, the mean paired change in overall $U_{\min}$ is $+0.114$ and the median is $+0.165$. Eight of twelve seeds move in a favourable direction. The 95% interval is $[-0.086, +0.294]$, however, so the direction is not stable across independently trained policies.

The success-only comparison is smaller. Mean success-only $\Delta U_{\min}$ is $+0.017$, with a 95% interval of $[-0.020, +0.062]$. Two seeds show clear within-success improvement: 900102 improves by $+0.215$ and 920106 by $+0.101$.

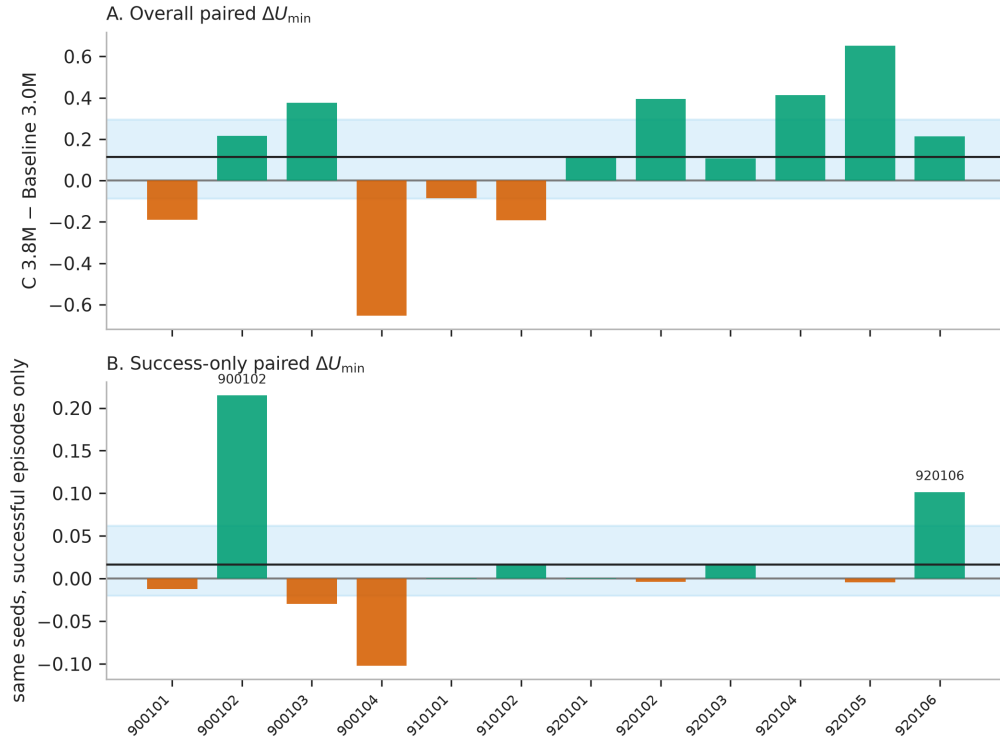


Figure 4.12.: Panel A shows seed-paired change in overall $U_{\min}$. Panel B shows the corresponding change within successful episodes only.

The two panels refine the favourable overall average. Four of the eight aggregate-

4. Experimental Development

positive seeds improve mainly through a better mix of successful and failed episodes rather than a clear redistribution among successful episodes. Overall $U_{\min}$ therefore provides evidence of a favourable endpoint shift, but it is not sufficient on its own to establish fairer sharing within successful coordination.

Supporting outcomes are directionally consistent with parts of this interpretation. Utility Gini, mean utility, completion, and collision move favourably on average, but their 95% intervals include zero. The Fast-minus-Slow burden gap narrows by -0.124 , with a 95% interval of $[-0.260, -0.015]$. Because this is a supporting outcome rather than the pre-specified primary outcome, it is used as evidence about the burden distribution rather than as a standalone validation of EXP05. Full values are reported in [section C.9](#).

4.6.6. Finding 2: The Local Burden Signal Reaches and Influences a Subset of Learned Policies

Across the full 3.0M-to-3.8M checkpoint trajectory, every seed activates the local rule at least once, so the mechanism is reachable during training. At the final endpoint the signal becomes sparse: on H1 at 3.8M, six of twelve seeds encounter no natural trigger, where a natural trigger means a normal decision reaches $g_i(t) > c$ and activates the local fairness penalty.

Policies that do encounter the signal vary in how strongly their actions depend on it. With the physical traffic state held fixed and only a calibrated neighbour-burden gap inserted, the selected action changes in 0.39%–4.04% of tested states across seeds. Some frequently triggered policies still move against the intended local incentive: seed 900101 selects **ACCELERATE** in all 845 triggered decisions, and seed 900103 accelerates in 96.6% of its triggered decisions.

Failure-episode coverage is limited as well. Ten final policies record collisions, yet the median share of collision episodes with no trigger is 99.6%. The mechanism reaches and

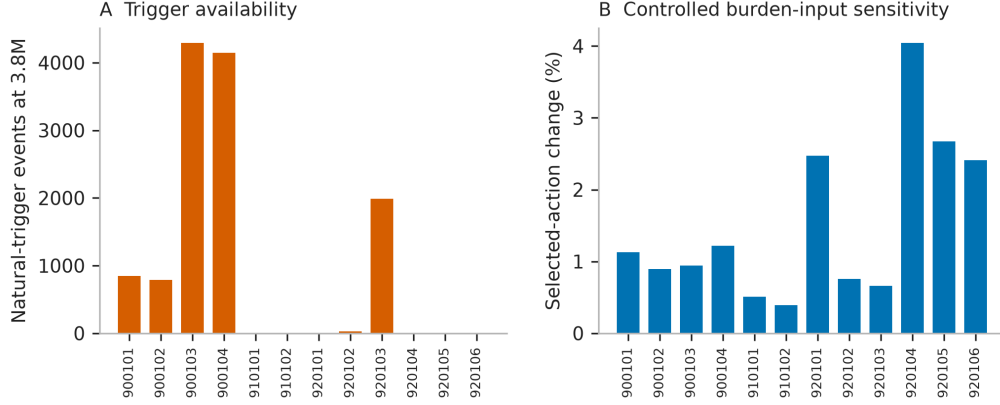


Figure 4.13.: Panel A shows natural-trigger availability by seed at 3.8M. Panel B shows selected-action change rates under the controlled neighbour-burden intervention.

influences some learned policies, but its final-state coverage and action sensitivity are not yet broad enough to explain the welfare outcome across seeds.

4.6.7. Finding 3: Training Produces Strong Intermediate Policies, but Their Behaviour Is Not Always Retained

The 17 saved checkpoints from 3.0M to 3.8M show that several policies pass through strong task regimes before the pre-specified endpoint. The changes between neighbouring checkpoints can nevertheless be large. Seed 900101 reaches 87.9% completion at 3.75M and falls to 50.4% at 3.80M; seed 900102 falls from 96.5% to 62.9% over the same interval. Seed 900104 reaches 95.7% completion at 3.70M, drops to 0.39% at 3.75M, and then partially recovers.

These intermediate checkpoints diagnose retention rather than select a more favourable result after training. Useful policies do emerge; the problem is that favourable behaviour can be transient and is not consistently retained at the final endpoint.

4. Experimental Development

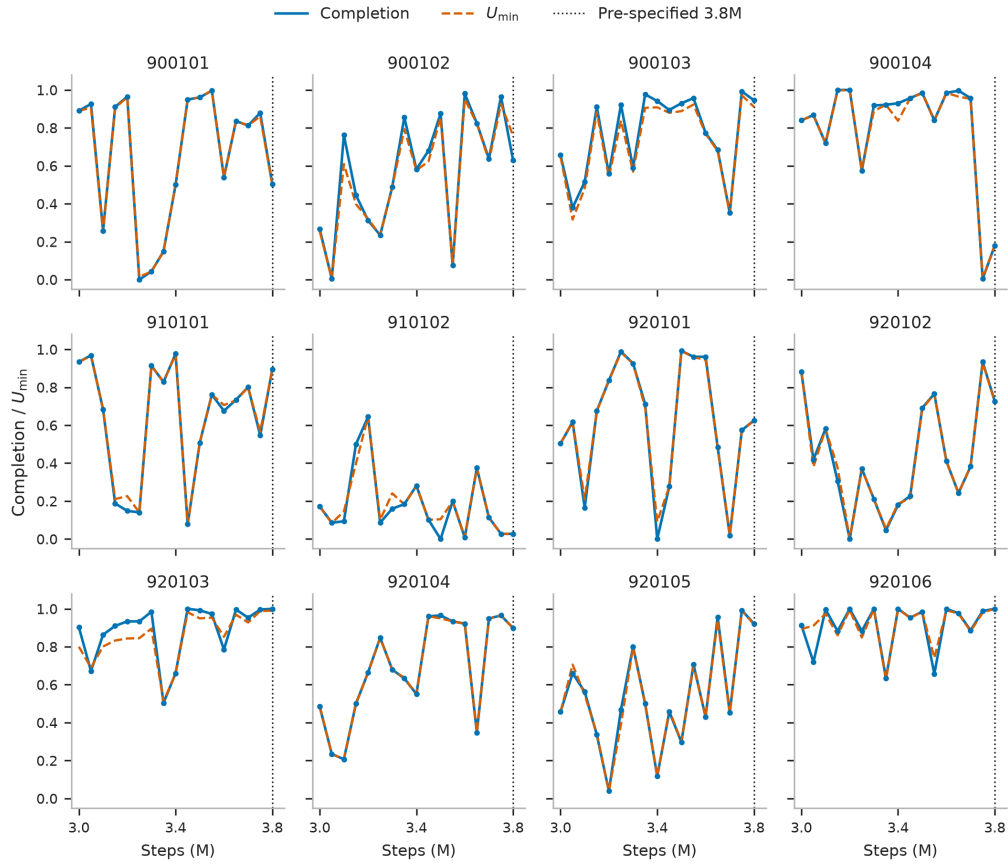


Figure 4.14.: Completion and worst-off utility across the 17 saved checkpoints from 3.0M to 3.8M. The pre-specified 3.8M endpoint is retained.

4.6.8. Overall Finding of EXP05

EXP05 offers positive but limited evidence. Overall worst-off welfare moves favourably on average, two policies show clear within-success gains, and the local mechanism is reachable and behaviourally relevant for a subset of policies. The effect is not seed-general, final trigger coverage is sparse, action sensitivity is often weak, and late training remains unstable.

The experiment supports the plausibility of a persistent, local, action-linked fairness mechanism without validating the combined package as a reliable fairness method. Because accumulated burden, neighbour comparison, and action-dependent feedback are introduced together, EXP05 cannot identify which component drives the favourable changes.

5. Discussion

[chapter 4](#) reports the experimental results. This chapter interprets them against the research questions and against the broader problem of turning a group-level welfare preference into local multi-agent behaviour.

5.1. Fairness under Heterogeneous Mobility Requirements

EXP01 shows why visible concession is not an adequate fairness measure in this environment. Slow vehicles brake more often, yet Fast vehicles sit more often in the welfare lower tail and carry greater target-relative mobility loss. When agents have different mobility requirements, equal-looking behaviour and equal experienced welfare are different questions.

The welfare definition plays the same role. Target-relative attainment is not offered as a universal measure of traffic fairness; it is a modelling choice matched to a task in which two classes have explicit, different target speeds. The contribution is methodological as well as empirical: a fairness analysis should specify the protected outcome before treating observable actions as evidence of disadvantage.

5.2. Why Social-Welfare Objectives Alone Were Insufficient

Mean, GGI, and Rawlsian Maximin say how to value a completed distribution of individual utilities. They do not say which vehicle should yield at a live merge conflict, whether the

5. Discussion

currently disadvantaged vehicle is visible, or whether a short-term concession will recover later.

EXP02 isolates that gap. The environment, task reward, observation, architecture, starting checkpoint, and continuation budget stay matched while only the terminal aggregation rule changes, yet no objective reliably improves $U_{\min}$ or Utility Gini across the 12 seeds. The result does not imply that social-welfare optimisation is ineffective in general. In this decentralised merge, the aggregation rule alone is too indirect to drive a stable local response.

Rawlsian Maximin makes the point especially clear. Failure-aware utility gives the terminal minimum a strong task-success component, so once an episode has succeeded the reward has relatively little variation left with which to distinguish successful welfare distributions. That helps explain why a worst-off objective can sound closely aligned with the research question while still exerting limited pressure on within-success redistribution.

5.3. Welfare History Can Support Local Behaviour

WSC separates information availability from the terminal objective. The live masking intervention shows that trained policies use the welfare-history channels: changing those inputs while holding the network and physical observations fixed changes episode outcomes. The targeted-yielding audit adds a more interpretable link—when a visible disadvantaged neighbour appears, the policy often concedes locally and welfare-gap recovery under WSC is substantially higher.

Final fairness still does not improve reliably. A representation can change local behaviour without producing stable trajectory-level redistribution, so mechanism evidence and endpoint evidence should not be collapsed into one success criterion.

5.4. From Temporal Credit to Local Credit Assignment

DWS changes when welfare feedback arrives but keeps the signal shared across active agents. Its diagnostics expose two separable problems. First, a short-horizon welfare change can misrepresent an action whose effect appears only later. Second, even timely feedback is hard to use when the same signal reaches agents that were not locally involved.

EXP05 responds by making disadvantage persistent through accumulated burden, comparing only visible neighbours, and tying local feedback to the ego action. The favourable mean endpoint shift and the two clear within-success improvements show that this package can coexist with useful fairness gains. The mechanism audits also show why those gains do not generalise across seeds: some final policies never encounter the trigger; some encounter it but rarely change action when burden information is manipulated; and most collision episodes lie outside the trigger’s coverage.

These results suggest a diagnostic sequence for future fair-MARL systems: does the policy encounter the fairness signal; does the signal change action selection; can that action influence the protected outcome; and is the behaviour retained during training? The present study finds failure at each link in at least some policies. It does not show that satisfying all four would be sufficient.

5.5. Overall Interpretation of the Research Questions

RQ1 is supported. Task-only coordination produces systematic distributional inequality, and the group identified by visible concession is not the group most exposed to requirement-relative welfare loss.

RQ2 is not supported in the strong form tested here. Mean, GGI, and Rawlsian Maximin do not reliably improve worst-off welfare or Utility Gini when only the terminal social-welfare objective changes while the rest of the learning setup stays matched. The

5. Discussion

later experiments refine rather than overturn that conclusion. WSC shows that explicit welfare history can become behaviourally useful; DWS shows that earlier feedback still faces temporal and local credit problems; and EXP05 shows that a more persistent, action-linked mechanism can produce favourable outcomes in some policies without yet becoming a stable, seed-general method.

6. Critical Evaluation and Future Work

This chapter steps back from the experimental narrative. I assess how far the project met its objectives, which design choices look sound in hindsight, and what a stronger follow-up study would change.

6.1. Achievement of the Project Aims and Objectives

The first objective was met clearly. EXP01 established and measured a distributional-fairness problem in a task that had already learned useful coordination, identified who is most exposed to welfare disadvantage, and showed why braking alone would have led to a misleading conclusion.

The second objective was deliberately harder: to test whether changing the social-welfare aggregation rule is enough to improve the realised distribution. The answer is negative but informative. The matched EXP02 design tests that claim directly and avoids attributing fairness changes to simultaneous architectural or observation changes.

The later work became diagnostic rather than confirmatory. WSC, DWS, and EXP05 locate concrete places where a group-level fairness preference can lose influence before it becomes stable local behaviour. They do not deliver a general solution, but they turn a null objective comparison into a set of testable mechanism questions.

6.2. Strengths of the Experimental Design

A major strength is the separation between physical concession and experienced welfare. Braking and similar actions are not treated as fairness outcomes unless they actually produce requirement-relative loss.

EXP02’s matched continuation is another. Within each seed, the welfare conditions start from the same 1.2M task checkpoint and keep the same environment, observation, network, and task reward. Training variability remains, but the comparison is substantially cleaner than training four unrelated systems from scratch.

The statistical unit also fits the training question. H1 contains 256 held-out scenarios per policy, yet episode outcomes are averaged within seed and uncertainty is estimated across the 12 independently trained policies. Hundreds of episodes from the same learned network are therefore not treated as independent training replications.

WSC adds an intervention test rather than relying on correlations alone. Replacing the welfare-history inputs after training changes outcomes without altering the physical observation or network weights, which directly tests whether the added representation is used.

6.3. Design Choices That Should Be Changed

6.3.1. Failure-Aware Utility Should Have Been Paired with Success-Only Welfare from the Start

Setting $U_i = 0$ after collision, timeout, or incompleteness gives task failure a real welfare cost, but it also merges two questions: whether the task succeeded and how welfare was distributed when it did. EXP02 and EXP05 both show that the split matters.

If I ran the study again, overall failure-aware welfare, success-only welfare, completion, and collision would be specified together before the formal experiments. The overall

metric would stay primary, but success-only welfare would stop safer or more successful execution from being mistaken for redistribution within successful merges.

6.3.2. DWS Was Too Coarse a Test of Earlier Fairness Feedback

DWS asks a sensible question—does a terminal welfare signal arrive too late to guide local actions?—but the implemented signal changes other properties as well. It compresses the welfare-history-floor change into $+0.0005$, 0 , or -0.0005 , discarding magnitude, and broadcasts the same value to every active vehicle.

The later audit shows why both choices matter. Short-horizon feedback can disagree with later recovery, and a substantial fraction of signal recipients are not locally involved in the event that produced the shared feedback. DWS therefore does not isolate timing cleanly; it mixes timing with sign-only compression and broad credit assignment.

A stronger design would compare an earlier shared signal with a matched local version while preserving more information about the size of the welfare change. That would separate “when did feedback arrive?” from “which agent should respond?”.

6.3.3. EXP05 Changed Too Many Components at Once

EXP05 replaces welfare history with accumulated burden, adds visible-neighbour burden comparison, and introduces an action-dependent local reward. The changes are motivated by earlier findings, but they arrive as one package.

The experiment can show that the package sometimes coexists with genuine within-success improvement, but it cannot show which component matters. Sparse triggers or low action sensitivity could reflect the burden representation, threshold calibration, the learned state distribution, or interactions between them.

A stronger final experiment would add a parallel 3.0M-to-3.8M continuation with the same terminal Rawlsian Maximin objective but without the new local mechanism. A small

6. Critical Evaluation and Future Work

component comparison could then separate burden observation, neighbour comparison, and action-dependent reward.

6.3.4. Stability Checks Should Have Been Planned Earlier

EXP05 uses a fixed 3.8M primary endpoint, yet the 17-checkpoint audit shows large swings over 50,000-step intervals. The endpoint should stay fixed to avoid post-hoc selection, but neighbouring checkpoints should have been in the evaluation plan from the start as a separate stability diagnostic.

That would preserve one actual saved policy as the object of inference while making policy retention an explicit property of training rather than an after-the-fact surprise.

6.4. Limitations and Threats to Validity

The environment is intentionally narrow: four learned vehicles, one merge geometry, two target-speed classes at 18 and 22 m/s, and one parameter-shared DQN. The control helps isolate the experiments but limits external validity. Denser traffic, other road layouts, heterogeneous vehicle dynamics, or non-shared policies could change both coordination and the welfare distribution.

Statistical uncertainty is estimated across 12 independent training seeds, and several confidence intervals remain wide. More seeds would improve precision without removing design confounds such as the bundled EXP05 intervention.

H1 is fixed, which strengthens matched comparisons because every policy sees the same held-out scenario set. The cost is narrower evidence for generalisation: the thesis does not test a new scenario distribution, different target-speed ratios, or larger numbers of controlled vehicles.

EXP05 carries the strongest internal-validity limitation because there is no parallel continuation control and three mechanism components change together. The post-hoc

mechanism audit helps locate where influence is lost, but it cannot establish that any one component is necessary, sufficient, or causal.

The welfare definition is task-specific as well. Target-relative mobility burden fits a merge with explicit speed requirements, but another traffic problem might protect travel time, safety exposure, access, comfort, or some combination instead.

6.5. Future Work

The next experiment should unpack what EXP05 currently bundles. Starting from the same 3.0M policies, a practical sequence would compare: a continuation control with terminal Rawlsian Maximin only; burden information without the local penalty; burden plus visible-neighbour comparison; and the full action-dependent mechanism. That would separate continued training from the intervention and provide a first component-level attribution.

Credit assignment is the second priority. Difference rewards and counterfactual multi-agent critics ask whether one agent's action changed the group outcome; temporal-credit methods address delayed consequences. Both are more complex than DWS, but they target the two weaknesses observed here: delayed recovery and diffuse responsibility.

The welfare definition also needs stress-testing. Follow-up studies could vary the Fast/Slow target-speed gap, increase the number of controlled vehicles, introduce non-shared or human-like traffic policies, and evaluate multiple road geometries and held-out scenario distributions. Those tests would show whether the target-relative findings persist beyond the controlled merge used in this dissertation.

7. Conclusion

This dissertation studies distributional fairness in a four-vehicle highway merge with heterogeneous mobility requirements.

7.1. RQ1 — Welfare Distribution under Baseline Coordination

RQ1 is supported. Baseline policies usually complete the merge, yet Fast vehicles appear disproportionately among worst-off outcomes and carry more target-relative mobility burden, while Slow vehicles brake more often. Physical concession and requirement-relative welfare therefore identify different groups.

7.2. RQ2 — Social-Welfare Objectives

The matched RQ2 comparison finds no reliable improvement from Mean, GGI, or Rawlsian Maximin when only the terminal social-welfare objective changes. Mean and GGI move favourably on average on some measures, but the effects vary across training runs; Rawlsian Maximin stays close to Baseline and adds little separation among successful episodes.

The diagnostic experiments clarify what happens between objective and behaviour. WSC shows that explicit welfare history can influence local decisions and targeted

7. Conclusion

welfare-gap recovery. DWS shows that earlier shared feedback still leaves temporal and local credit problems. EXP05 shows that a persistent burden representation with local, action-linked feedback can produce favourable outcomes in some learned policies, but not yet with seed-general reliability.

7.3. Overall Contribution

I keep the main empirical conclusions narrow. Visible concession and experienced mobility burden can disagree; changing the social-welfare objective alone does not reliably improve final fairness; and welfare information can alter local behaviour without guaranteeing a stable change in the final distribution. The mechanism audits turn those results into concrete design questions about signal availability, action sensitivity, outcome coverage, and training stability.

A. AI-Assisted Work

AI tools were used as supporting tools for grammar and language checks, structural editing, and debugging code during development and dissertation preparation. This included identifying unclear wording, checking grammar and consistency, suggesting section reorganisation and shorter figure/table captions, and helping trace or explain coding errors.

The research design, experimental decisions, empirical analysis, interpretation of results, and acceptance or rejection of editorial suggestions remain the author's responsibility. AI output was not treated as empirical evidence, and reported results were checked against the project code, logs, analysis outputs, and cited sources.

B. Curriculum Scenario Bank and Reproducibility

This appendix records the rules used to construct the fixed curriculum scenario bank and the main implementation settings needed to reproduce the formal study. The complete 64-scenario bank is kept in the frozen project scenario files rather than reproduced row by row here.

B.1. Scenario Bank Construction

Road geometry is shown in [fig. 3.1](#). The curriculum contains 64 initial traffic configurations (Q_00000–Q_00063) of the same four-vehicle merge task. Every scenario contains one Fast and one Slow vehicle on the ramp and one Fast and one Slow vehicle on the mainline.

Fast vehicles always start at their 22 m/s target speed and Slow vehicles at their 18 m/s target speed. Each vehicle is assigned to one of two nominal arrival waves, centred at 4.5 s and 6.5 s from the conflict region. A shared scenario jitter of $\epsilon \sim \mathcal{U}(-0.25, 0.25)$ s and a smaller per-vehicle jitter of $\zeta_i \sim \mathcal{U}(-0.05, 0.05)$ s vary the exact arrival times. Scenarios are rejected if two vehicles in the same lane would start less than 15 m apart.

The role-speed class attached to physical vehicle IDs (V0–V3) is randomised by the scenario generator. The randomisation is reproducible from the stored episode seed. These 64 configurations are training inputs, not 64 different tasks.

B.2. Stored Initial Conditions

For each vehicle, the frozen scenario file stores its road role, speed class, target speed, spawn position, and nominal time-to-conflict. Spawn position and nominal time-to-conflict follow $\text{route_position} = \text{merge_start} - v_i^{\text{target}} \times \text{nominal_ttc}$. The full per-scenario values are retained in the project data used for training. Reprinting all 64 rows in the dissertation would not add information beyond the construction rules above.

B.3. Training, Observation, and Reward Details

B.3.1. Curriculum schedule

Stage	Scenarios	Training steps	Cumulative steps
C1	1	400,000	400,000
C4	4	300,000	700,000
C16	16	250,000	950,000
C64	64	250,000	1,200,000

Table B.1.: Curriculum task-learning schedule.

B.3.2. WSC observation structure

Block	Features
Ego	road role, speed, target speed, acceleration, distance to merge, previous action, $M_i(t)$
Visible neighbour	presence, relative distance, relative speed, lane relation, $M_j(t)$

Table B.2.: WSC welfare-history observation structure.

B.3. Training, Observation, and Reward Details

Component	Coefficient
Route progress	+0.4
Completion	+0.6
Controlled-vehicle collision	−1.0
Hard braking	−0.015
Timeout	−0.6

Table B.3.: Baseline task-reward components.

B.3.3. Baseline task reward

B.3.4. Core DQN settings

Parameter	Value
Observation dimension	18 (Original); 22 (WSC / burden-augmented)
Hidden layers	64, 64
Activation	ReLU
Actions	3
Optimiser	Adam [37]
Initial learning rate	5×10^{-4}
Final learning rate in original formal continuation	1×10^{-4}
Replay buffer	100,000 transitions
Batch size	64
Discount factor γ	0.995
TD return	1-step
Target	Double DQN [35]
Target-network update	Every 250 gradient updates
Initial exploration ϵ	1.0
Final exploration in original formal continuation	0.10

Table B.4.: Core DQN architecture and learning parameters for the original formal experiments.

For $N = 4$ controlled vehicles, the Baseline task reward is $r_t^{\text{task}} = 0.4 \left(\frac{1}{N} \sum_i \Delta \rho_{i,t} \right) + 0.6 \left(\frac{1}{N} \sum_i X_{i,t+1} \right) - C_{t+1}^{\mathcal{V}} - 0.015 \left(\frac{1}{N} \sum_i H_{i,t+1} \right) - 0.6 I_{t+1}^{\text{timeout}}$. Here, $\Delta \rho_{i,t}$ is route progress, $X_{i,t+1}$ completion, $C_{t+1}^{\mathcal{V}}$ controlled-vehicle collision, $H_{i,t+1}$ hard braking, and I_{t+1}^{timeout} timeout.

B. Curriculum Scenario Bank and Reproducibility

B.3.5. Longitudinal controller

The discrete action changes the desired-speed setpoint. The low-level controller requests $a_t^{\text{requested}} = K_P(v_t^{\text{desired}} - v_t)$, with $K_P = 1/0.6 \approx 1.67 \text{ s}^{-1}$ and acceleration clipped to $-3 \leq a_t \leq 2 \text{ m/s}^2$.

The author-written implementations of individual utility, the three social-welfare objectives, WSC, DWS, and the EXP05 local reward are listed in [chapter E](#). Simulator and standard DQN library code is not reproduced.

C. Robustness, Endpoint, and Training Diagnostics

This appendix keeps technical checks that are important for reproducibility but would interrupt the main experimental story if reported in full in [chapter 4](#).

C.1. Historical Late-Checkpoint Evaluation Audit

The original 2.0M analysis combined four saved networks from 1.85M to 2.00M. For each decision, their action scores were averaged before one action was selected. This sometimes reduced seed-level variation, but the four networks did not always recommend the same action: agreement across all four checkpoints was only 14.6% in the audit.

The DWS result is the main reason this matters. Under the four-checkpoint ensemble, Original+DWS gives approximately $\Delta U_{\min} = -0.1395$ and $\Delta G = +0.0735$. Evaluating the single 2.0M network instead gives approximately $+0.0663$ and -0.0243 .

DWS comparison	$\Delta U_{\min}$	Δ Utility Gini
Four-checkpoint ensemble	-0.1395	+0.0735
Single 2.0M checkpoint	+0.0663	-0.0243

Table C.1.: Endpoint sensitivity of Original+DWS. The two rows use the same H1 test set and differ only in how the late policy is represented.

This historical audit is retained for transparency because the original analysis used

checkpoint averaging. The main text treats the actual 2.0M DWS network as the policy object of interest and does not introduce the ensemble as a primary experimental condition. EXP05 uses the single pre-specified 3.8M network.

C.2. Why 3.0M Was Used before EXP05

The four development seeds had broadly consolidated task performance by 3.0M, although their saved policies still changed between nearby checkpoints. Extending task-only training to 3.5M did not give a consistently better starting point.

Seed	3.5M completion	3.5M collision
900101	0.727	0.266
900102	0.000	1.000
900103	0.680	0.313
900104	0.898	0.102

Table C.2.: Task performance of the four development seeds at 3.5M.

The 3.5M extension is mixed and includes a complete collapse for 900102. This is why 3.0M was used as the EXP05 starting point. It is described as a late task checkpoint, not as a converged policy.

C.3. Seed-Level Sensitivity and Technical Validity

Formal conclusions use the 12 training seeds as the independent replications. Sensitivity checks sometimes repeat an analysis after removing one seed at a time, but this is used only to show whether one training run has a large influence. It is not used to remove inconvenient policies from the main result.

A poor-performing run remains in the analysis if it completed the frozen training protocol, contains the required saved networks, and produces valid evaluation outputs.

Formal implementation checks also verify the WSC network expansion, DWS reward path, fixed scenario sets, and deterministic held-out evaluation.

C.4. Reproducibility Outputs

For every final figure, the repository should contain the source CSV/JSON, the script used to create the figure, and the exact checkpoint and scenario identifiers used. Learning curves in [chapter 4](#) start at training step 0, and endpoint figures use seed-level summaries whenever the training seed is the replication unit.

EXP05 should additionally preserve the calibrated c and g_{95} values, FULL/MASKED evaluation outputs, natural-trigger logs, controlled burden-input perturbations, collision-trigger alignment outputs, and the complete 3.0M–3.8M 50k-checkpoint trajectory.

C.5. EXP01 Supporting Diagnostics

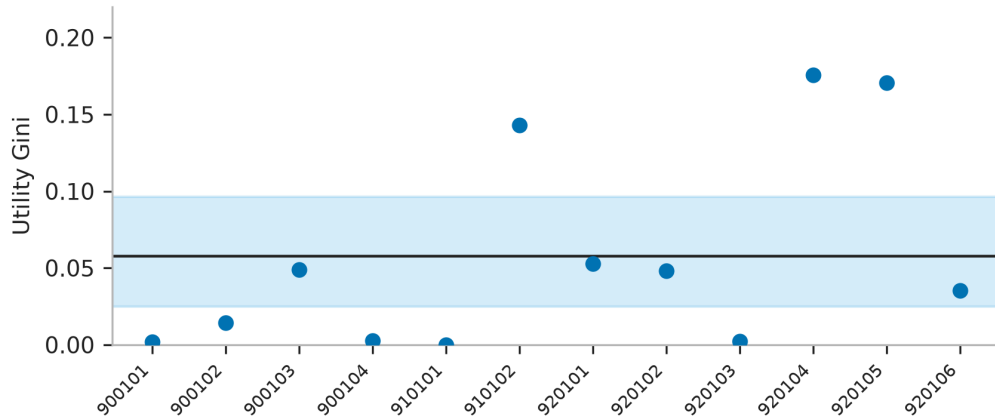


Figure C.1.: Utility Gini for each Baseline training seed. The horizontal line shows the across-seed mean and the shaded region its seed-level bootstrap confidence interval.

C. Robustness, Endpoint, and Training Diagnostics

Behaviour or burden measure	Slow	Fast
BRAKE rate	8.94%	5.52%
$P(v \geq v^{\text{target}} \mid \text{BRAKE})$	97.8%	85.0%
BRAKE while $u = 1$	96.7%	83.7%
Mean $v - v^{\text{target}}$ during BRAKE	+7.38 m/s	+3.83 m/s
Active time below target	1.37%	3.26%
Relative shortfall when below target	0.087	0.135

Table C.3.: Detailed physical-concession and requirement-relative mobility measures under the Baseline.

C.6. EXP02 Supporting Diagnostics

Task performance is retained here as context because final utility is failure-aware.

Condition	Completion	Collision	Timeout
Baseline	0.884	0.116	0.000
Mean	0.903	0.097	0.000
GGI	0.895	0.098	0.007
Rawlsian Maximin	0.872	0.108	0.020

Table C.4.: Held-out task performance for the matched EXP02 conditions.

The main text uses seed-paired changes because each welfare continuation is matched to the Baseline from the same training seed. Absolute endpoint summaries are retained here for context.

Condition	$U_{\min}$	Utility Gini	Mean U	Utility range
Baseline	0.882	0.058	0.941	0.118
Mean	0.902	0.049	0.951	0.098
GGI	0.895	0.050	0.944	0.100
Rawlsian Maximin	0.874	0.058	0.938	0.125

Table C.5.: Absolute EXP02 welfare outcomes. These values provide context but do not replace the paired seed-level comparison in [fig. 4.6](#).

The full EXP02 learning curve is reported in [fig. 4.5](#), starting from training step 0.

Additional Rawlsian Maximin checks support the narrow interpretation in [section 4.3.4](#). Median successful-episode $U_{\min}$ is 1.0 under both Baseline and Rawlsian Maximin; 93.2% and 89.6% of successful episodes respectively have $U_{\min} > 0.98$; the success-only paired change is approximately -0.010 with 95% interval $[-0.028, +0.002]$; and the worst-off identity changes about once per episode on average. The switching result is retained as a supporting check rather than a main explanation.

C.7. WSC Supporting Diagnostics

The matched interaction estimates used in [section 4.4.2](#) are:

Reward	Extra WSC effect on $U_{\min}$	95% CI	Extra WSC effect on Utility Gini	95% CI
Mean	-0.049	$[-0.236, +0.175]$	+0.023	$[-0.089, +0.118]$
GGI	+0.074	$[-0.037, +0.209]$	-0.038	$[-0.102, +0.016]$
Rawlsian Maximin	-0.018	$[-0.152, +0.097]$	+0.013	$[-0.044, +0.079]$

Table C.6.: WSC interaction estimates. Positive $U_{\min}$ and negative Utility Gini are favourable; all intervals include zero.

Fixed-state welfare-input interventions give the following selected-action change rates:

Welfare-input intervention	Selected-action change
Set all welfare values to 1	4.94%
Replace welfare values with the seed mean	6.64%
Permute welfare values	4.22%
Swap ego and neighbour-0 welfare	0.19%
Reduce neighbour-0 welfare by 0.3	2.83%
Increase neighbour-0 welfare by 0.3	0.10%

Table C.7.: Supporting WSC input-sensitivity checks with the physical observation held fixed.

The stronger response to an artificial worsening of a neighbour is descriptive evidence

C. Robustness, Endpoint, and Training Diagnostics

that the welfare channels affect the action rule. Live masking provides the main-text intervention result because it tests the same issue over complete episodes. A live permutation produced a small favourable aggregate change, but no mechanism was established for that result, so it is not used to explain WSC.

C.8. DWS Supporting Diagnostics

The main text uses three DWS checks: how closely the running signal follows final welfare, how long recovery can take after a local action, and whether the shared signal reaches agents that were not locally involved.

Diagnostic	Estimate
Correlation between episode-level running-floor summary and final $U_{\min}$	$r \approx 0.727$
More positive than negative DWS feedback but final $U_{\min} < 0.5$	8.3%
Recovery at +5 steps (Original / WSC)	0.1% / 3.1%
Recovery at +50 steps (Original / WSC)	57.4% / 65.6%
Recipients not locally involved (Original+DWS / WSC+DWS)	41.7% / 24.4%

Table C.8.: Supporting DWS diagnostics used in [section 4.5.3](#).

C.9. EXP05 Supporting Diagnostics

EXP05 continuation uses learning rate 5×10^{-5} , exploration 0.02, and gradient-norm clipping at 10. The exact frozen calibration values are $c = 0.007784310312400124$, $g_{95} = 0.027956245271296754$, and $\eta_{\max} = 0.0015$. They were calculated from the four development seeds 900101–900104 at the 3.0M task base and were not re-selected from the 12-seed primary outcome.

Additional paired outcomes are timeout +0.002 with 95% interval $[-0.009, +0.016]$, mean cumulative burden -0.187 with interval $[-0.659, +0.219]$, and worst-off burden -0.426 with interval $[-1.473, +0.540]$. The Fast-minus-Slow burden-gap change is -0.124

Outcome	Mean paired change	95% CI	Favourable seeds
Utility Gini	-0.053	$[-0.143, +0.047]$	8/12
Mean utility	+0.062	$[-0.043, +0.155]$	8/12
Completion	+0.092	$[-0.109, +0.278]$	7/12
Collision	-0.094	$[-0.276, +0.097]$	7/12

Table C.9.: Supporting seed-paired changes from the 3.0M Baseline to the EXP05 policy at 3.8M.

with interval $[-0.260, -0.015]$.

Across the 17 saved checkpoints from 3.0M to 3.8M, every seed activates the local rule at least once, but activation becomes sparse near the final endpoint. At 3.8M, six of twelve seeds have no natural trigger.

To test whether the trained action rule responds to burden information, the physical state is held fixed while a large calibrated neighbour-burden gap is inserted. The selected action changes in only 0.39%–4.04% of tested states across seeds. The two clear within-success improvement seeds are not unusually sensitive (900102: 0.89%; 920106: 2.41%).

Among naturally triggered decisions at 3.8M, 900101 selects **ACCELERATE** on 845/845 events and 900103 does so on 96.6% of its events. Ten seeds have collision episodes at 3.8M; the median fraction of those collision episodes with no trigger is 0.996. These checks support [section 4.6.6](#) but do not identify one causal failure mode.

The full 3.0M–3.8M checkpoint trajectory is reported in [fig. 4.14](#) rather than duplicated in the appendix.

D. Earlier Development Experiments

The formal conclusions in Chapters 4–7 come only from the four-vehicle highway-merging study. Earlier traffic experiments are summarised here only to show how the final task was selected and developed. Their environments, rewards, agent counts, and evaluation procedures differ, so they are not statistical baselines or evidence that the final fairness results generalise.

Earlier experiment	Main observation	Relevance to the final study
Two-vehicle head-on meeting	The same-speed task was solved reliably across 10 seeds; a heterogeneous-speed version still achieved about 0.955 safe passing on average	Showed that learned coordination can remain feasible when agents have different mobility goals
Five-vehicle overtaking	Across 9 complete seeds, final completion was about 0.067 and collision about 0.933	Showed that some traffic tasks were dominated by basic task failure and were poor choices for a fairness study
Earlier two- and four-vehicle merge pilots	Several reward and network designs achieved useful task competence, but protocols and welfare definitions differed from the final study	Helped refine the merge geometry, agent count, and training setup

Table D.1.: Development experiments retained for context only.

The head-on meeting task showed that a second traffic geometry could support stable learned coordination, including with heterogeneous speed goals. The overtaking task showed the opposite problem: when collisions dominate, changes in welfare are difficult

D. Earlier Development Experiments

to separate from failure to learn the task itself. Earlier merge pilots then helped narrow the final environment and controller design.

No Mean, GGI, Rawlsian Maximin, WSC, or DWS conclusion in the thesis is based on these earlier experiments. Their role is developmental rather than evidential.

E. Essential Implementation

This appendix lists the author’s own implementations of the definitions used in Chapters 3 and 4. The simulator is `highway-env` 1.12.0 [33]. The controller is a standard parameter-shared Double DQN [34], [35] trained with Adam [37]. Those libraries, the training loop, and analysis scripts are not reproduced.

The listings are shortened only by removing comments and unused helpers. The executable statements are unchanged. Formal welfare-objective training passed `lam=0.5` into `terminal_welfare_bonus`, matching $R_c^W = 0.5[W_c(\mathbf{U}) - 1]$ in section 3.7.3; the module default `WELFARE_LAMBDA=1.0` was not used for those runs.

E.1. Individual Welfare and Burden

`target_speed_attainment` and `episode_mobility_outcome` implement $u_{i,t}$ and failure-aware U_i (section 3.6.1). `coordination_burden` implements C_i (section 3.6.2). `running_active_attainment` is the implementation function that computes the live-prefix active-attainment mean used as the welfare-history variable $M_i(t)$ in WSC.

```
# thesis/pilots/stage11_welfare.py
def target_speed_attainment(speed: float, target_speed: float) -> float:
    if target_speed <= 0:
        raise ValueError(f"target_speed must be > 0, got {target_speed}")
    return float(np.clip(speed / target_speed, 0.0, 1.0))

def mean_welfare(experiences: Sequence[float]) -> float:
    if not experiences:
        raise ValueError("experiences must be non-empty")
    return float(sum(experiences) / len(experiences))
```

E. Essential Implementation

```
def min_welfare(experiences: Sequence[float]) -> float:
    if not experiences:
        raise ValueError("experiences must be non-empty")
    return float(min(experiences))

def episode_mobility_outcome(*, collided: bool, attainment_samples: Sequence[float]) -> float:
    if collided:
        return 0.0
    if not attainment_samples:
        return 0.0
    return float(sum(attainment_samples) / len(attainment_samples))
```

```
# thesis/study_b/utility.py
GGI_WEIGHTS_N4: tuple[float, ...] = (0.4, 0.3, 0.2, 0.1)

def gini_coefficient(values: Sequence[float]) -> float | None:
    n = len(values)
    if n == 0:
        raise ValueError("values must be non-empty")
    total = sum(values)
    if total == 0:
        return None
    numerator = sum(abs(a - b) for a in values for b in values)
    return float(numerator / (2.0 * n * total))

def coordination_burden(
    *, dt: float, active_flags: Sequence[bool], attainments: Sequence[float]
) -> float:
    if len(active_flags) != len(attainments):
        raise ValueError("active_flags and attainments must be the same length")
    return float(dt * sum((1.0 - u) for active, u in zip(active_flags, attainments) if active))

def episode_utilities(traces: Mapping[str, EpisodeVehicleTrace]) -> dict[str, float]:
    out: dict[str, float] = {}
    for vid, trace in traces.items():
        active_attainments = [
            a for active, a in zip(trace.active_flags, trace.attainments()) if active
        ]
        out[vid] = episode_mobility_outcome(
            collided=trace.collided, attainment_samples=active_attainments
        )
    return out

def running_active_attainment(trace: EpisodeVehicleTrace) -> float:
    active_samples = [
        a for active, a in zip(trace.active_flags, trace.attainments()) if active
    ]
    if not active_samples:
        return 1.0
    return float(sum(active_samples) / len(active_samples))
```

E.2. Social-Welfare Objectives and Terminal Bonus

GGI sorts utilities ascending and applies the frozen weights from [section 3.7.2](#). The terminal bonus is computed once per episode and added to every agent's own terminal task reward.

```
# thesis/study_b/utility.py
def generalized_gini_welfare(
    values: Sequence[float], weights: Sequence[float] = GGI_WEIGHTS_N4
) -> float:
    n = len(values)
    if n == 0:
        raise ValueError("values must be non-empty")
    if len(weights) != n:
        raise ValueError(f"weights length {len(weights)} must match values length {n}")
    ordered = sorted(values)
    return float(sum(w * u for w, u in zip(weights, ordered)))
```

```
# thesis/study_b/welfare_reward.py
WELFARE_LAMBDA: float = 1.0 # module default; formal runs passed lam=0.5

@dataclass(frozen=True)
class WelfareCondition:
    name: str
    welfare_fn: Callable[[Sequence[float]], float]

MEAN = WelfareCondition("mean", mean_welfare)
GGI = WelfareCondition("ggi", generalized_gini_welfare)
MAXIMIN = WelfareCondition("maximin", min_welfare)

def terminal_welfare_bonus(
    condition: WelfareCondition,
    episode_utilities: Sequence[float],
    *,
    lam: float = WELFARE_LAMBDA,
) -> float:
    if not episode_utilities:
        raise ValueError("episode_utilities must be non-empty")
    w_c = condition.welfare_fn(list(episode_utilities))
    return float(lam * (w_c - 1.0))
```

E.3. WSC Observation Features

WSC appends $M_i(t)$ to the ego block and $M_j(t)$ to each of the same three neighbour slots, expanding the observation from 18 to 22 dimensions (section 3.4.2). Neighbour target speeds are not included.

```
# thesis/study_b/local_observation.py
SELF_OBS_DIM: int = 6
NEIGHBOUR_OBS_DIM: int = 4
LOCAL_OBS_DIM: int = SELF_OBS_DIM + NEIGHBOUR_SLOTS * NEIGHBOUR_OBS_DIM # 18
SELF_OBS_DIM_WSC: int = SELF_OBS_DIM + 1 # 7
NEIGHBOUR_OBS_DIM_WSC: int = NEIGHBOUR_OBS_DIM + 1 # 5
LOCAL_OBS_DIM_WSC: int = SELF_OBS_DIM_WSC + NEIGHBOUR_SLOTS * NEIGHBOUR_OBS_DIM_WSC # 22

def build_self_observation(
    ego: VehicleSnapshot, *, merge_start: float, include_welfare_state: bool = False
) -> np.ndarray:
    fields = [
        encode_role(ego.role),
        float(ego.speed),
        float(ego.target_speed),
        float(ego.acceleration),
        distance_to_merge(ego.route_position, merge_start=merge_start),
        encode_peer_intent(int(ego.previous_action)),
    ]
    if include_welfare_state:
        fields.append(float(ego.welfare_state))
    return np.array(fields, dtype=np.float64)

# thesis/study_b/local_observation.py (neighbour row after slot selection)
row = [1.0, delta_d_norm, delta_v_norm, lane_relation]
if include_welfare_state:
    row.append(float(other.welfare_state))
```

E.4. Dense Welfare Shaping

DWS evaluates the same welfare function on the current welfare-history values $M_i(t)$ and adds a shared discrete term F_t when $\Delta\Phi_t$ exceeds a frozen threshold (section 4.5.1). The snapshot uses a fixed vehicle-id set so that a vehicle leaving the road cannot create a spurious welfare gain.

```
# thesis/study_b/dense_shaping.py
```

```

def welfare_objective_snapshot(
    traces: Mapping[str, EpisodeVehicleTrace],
    vehicle_ids: Sequence[str],
    condition: WelfareCondition,
) -> float:
    m_values = [running_active_attainment(traces[vid]) for vid in vehicle_ids]
    return condition.welfare_fn(m_values)

def dense_shaping_term(delta_phi: float, config: DenseShapingConfig) -> float:
    if not config.enabled:
        return 0.0
    if delta_phi > config.epsilon:
        return float(config.magnitude)
    if delta_phi < -config.epsilon:
        return float(-config.magnitude)
    return 0.0

```

E.5. EXP05 Local Fairness Reward

The follow-up reuses C_i as accumulated burden, compares the ego vehicle only with visible neighbours, and turns a positive local gap into an action-dependent penalty ([section 4.6](#)). BRAKE, HOLD, and ACCELERATE map to weights 0, 0.5, and 1.

```

# mechanism_c_v2_stabilized/scripts/mechanism_c_core.py
ACTION_WEIGHTS: dict[int, float] = {BRAKE: 0.0, HOLD: 0.5, ACCELERATE: 1.0}
ETA_MAX: float = 0.0015

def cumulative_burden(trace: EpisodeVehicleTrace, *, dt: float) -> float:
    return coordination_burden(
        dt=dt, active_flags=trace.active_flags, attainments=trace.attainments()
    )

def normalized_burden(cumulative: float, *, c_max: float) -> float:
    if c_max <= 0:
        raise ValueError(f"c_max must be > 0, got {c_max}")
    return float(np.clip(cumulative / c_max, 0.0, 1.0))

def local_inequity_gap(
    ego_burden: float,
    neighbour_burdens: Sequence[float],
    neighbour_presence: Sequence[float],
) -> float:
    if len(neighbour_burdens) != len(neighbour_presence):
        raise ValueError("neighbour_burdens and neighbour_presence must be the same length")
    visible_gaps = [
        nb - ego_burden

```

E. Essential Implementation

```
        for nb, p in zip(neighbour_burdens, neighbour_presence)
            if p > 0.5
        ]
    if not visible_gaps:
        return 0.0
    return float(max(visible_gaps))

def inequity_severity(g: float, *, c: float, g95: float) -> float:
    if g95 <= c:
        raise ValueError(f"g95={g95} must be > c={c}")
    excess = max(g - c, 0.0)
    return float(np.clip(excess / (g95 - c), 0.0, 1.0))

def local_fairness_reward(
    g: float, action: int, *, c: float, g95: float, eta_max: float = ETA_MAX
) -> float:
    if g <= c:
        return 0.0
    severity = inequity_severity(g, c=c, g95=g95)
    return float(-eta_max * severity * ACTION_WEIGHTS[action])
```

These listings are the essential parts of the implementation. Printed at reduced size (two logical pages per sheet) they occupy at most five physical pages, as required for an implementation project.

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